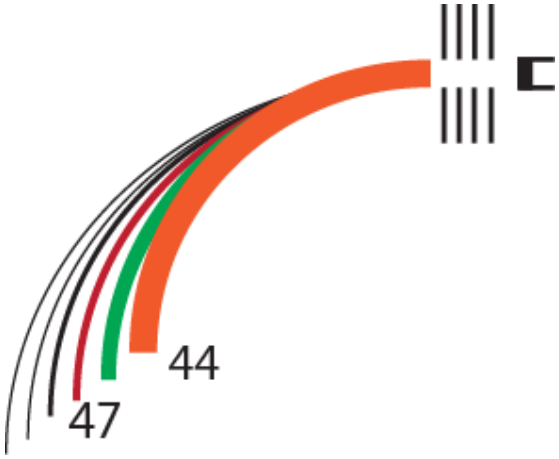
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Variable Δ_{47} acid digestion fractionation

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The extraction of CO_2 from $CaCO_3$ via digestion by phosphoric acid includes a fractionation factor that affects both $\delta^{18}O$ and Δ_{47}. In both systems the fractionation factor is assumed to vary only with temperature and to be constant at a specific temperature. A correction factor is therefore derived to enable comparison of data that is produced under different acid digestion temperatures, reflecting the difference between 25°C reaction and a reaction at a higher temperature ($\Delta\delta_{25\text{-}T}$). Calcite samples having a wide range of Δ_{47} values (0.26‰ - 0.74‰) and of bulk composition ($\delta47$ values between -40‰ and +20‰) were analyzed using acid digestion at either 25°C or 90°C. The difference in Δ_{47} values when digesting at 90°C versus 25°C ($\Delta\delta_{25\text{-}90}$) is highly variable, suggesting that the acid digestion fractionation is not constant at a given temperature. $\Delta\delta_{25\text{-}90}$ varies between the accepted value of 0.09‰ to as low as 0.03‰, and correlates with the bulk isotopic composition of the sample (either $\delta47$ of the sum of $\delta^{13}C+\delta^{18}O$). This strong variability is observed in particular when phosphoric acid concentration is at the low end of commonly used concentrations (103%); weaker variability is observed when using higher acid concentration (105%). Such non-constant $\Delta\delta_{25\text{-}90}$ leads to potential systematic errors in Δ_{47} determination in experimental setups in which acid concentration is relatively low, or if concentrations change over time, as may often be the case in a common acid bath.

How small can we go? First results of a Thermo Scientific 253-Plus with a Kiel IV device for carbonate clumped isotopes

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One of the main factors still limiting the application of carbonate clumped isotope thermometry to many fields of research is the relatively large amount of sample necessary for accurate measurements. In systems using a common acid bath preparation system, the most commonly used sample digestion methodology, a single analysis requires 5 to 10 mg of sample (e.g. Passey et al., 2010). Considering that for a precise temperature estimate, 3 replicate analyses are desirable, the total sample amounts required is in the range 15 to 30 mg. With the use of a Kiel IV coupled to a MAT 253, we have demonstrated that a long-term external reproducibility in Δ_{47} of 0.011-0.016 ‰ (1 SD) can be achieved with the sample sizes of 1.5 to 2 mg, amounting to 4.5 to 6 mg total sample for a triplicate measurement (Meckler et al., 2014). This sample reduction, coupled with the fact that samples are repeatedly measured in small aliquots of 150-200 μg, is opening new venues of research. For example, with this technique the analysis of foraminifera becomes less time-consuming and, in addition, it is possible e.g. to produce high-resolution oxygen and carbon isotope curves contemporaneously with a low-resolution clumped isotope temperature reconstruction (Grauel et al., 2013).

In this contribution we will present the first data obtained with the new Thermo Scientific 253 Plus coupled to a Kiel IV device. The improved collector design has dramatically reduced negative backgrounds on the *m/z* 47-49 collectors and allows simultaneous baseline monitoring during sample measurement, simplifying the correction procedures. In addition, we demonstrate the reproducibility and precision achievable with the implementation of the new LIDI measurement workflow (Hu et al., 2014) in the Isodat software.

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Passey, B. H., Levin, N. E., Cerling, T. E., Brown, F. H., and Eiler, J. M., 2010, High-temperature environments of human evolution in East Africa based on bond ordering in paleosol carbonates: Proceedings of the National Academy of Sciences of the United States of America, v. 107, no. 25, p. 11245-11249.

Combining Mg/Ca and clumped isotope analyses for temperature estimation from foraminifera

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Reconstructing past temperature beyond the instrumental record is challenging and paleotemperature proxies measured on different phases often disagree (e.g., alkenone SST and foraminifer Mg/Ca). We combined clumped isotopes and Mg/Ca thermometry on the same specimens of different Atlantic planktic foraminifera to provide independent estimates of temperature as well as the $\delta^{18}O$ of the ambient seawater. The strength of the tandem approach is that unlike other palaeotemperature methods, Mg/Ca and Δ_{47} are measured on the exact same calcite phase, thereby eliminating differences in seasonality or depth habitat of proxies. This redundancy constitutes a rigorous test of individual methods with the advantage that the same approach can be applied to fossil specimens.

Aliquots for clumped and Mg/Ca analyses are treated in exactly the same way following a modified procedure after Barker et al. [2003], including a larger volume of reactant, stronger oxidative cleaning at room temperature for twice 10 minutes to remove contaminants from foraminiferal tests. Tests were gently crushed and 6-10 aliquots of 135-148 ug were transferred into KIEL glass vials for clumped analysis and another aliquot of 100-300 ug was used for Mg/Ca analysis by ICP-OES.

Clumped isotope analysis is carried out on a KIEL IV coupled to a Thermo Fisher MAT 253 mass spectrometer and following the procedure described in Meckler et al. [2014] and Kele et al. [2015]. As all cups on our instrument have the same width, backgrounds are monitored using cup30 (m/z 46.5), conveniently located in the middle of the cup array. This allows rapid assessment of machine and run performance with regard to contamination and mass spectrometric effects. For evaluation we use the carbonate standards ETH1, 2, 3, 4 and CarZ. Data are background, pressure baseline and carbonate standard corrected and reported in the absolute reference frame [*Kele et al*. 2014].

Preliminary data confirm the theoretical Mg/Ca-Δ_{47} relationship and hence corroborate independently the validity of the tandem method. This redundancy strengthens confidence in both methods and permits application of the same approach to fossil samples.

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Kele S. et al. (2015) Temperature dependence of oxygen- and clumped isotope fractionation in carbonates: A study of travertines and tufas in the 6–95°C temperature range. *Geochimica et Cosmochimica Acta* 168, 172–192.
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Effects of water on carbonate clumped isotope bond reordering kinetics

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Carbonate clumped isotope geothermometry is a powerful tool for reconstructing past temperatures. The overabundance of doubly substituted isotopologues can change at elevated temperatures (ca. > 100 °C) without altering the individual C and O bulk isotopic composition of a mineral owing to solid-state reordering reactions through the crystal lattice [*Dennis and Schrag*, 2010; *Passey and Henkes*, 2012]. Understanding the kinetics of this clumped isotope reordering process is a prerequisite for application to geological questions involving samples that have been heated in the subsurface.

The effect of water on reordering kinetics has not been thoroughly explored. The presence of water dramatically increases rates of oxygen self-diffusion in calcite [*Farver*, 1994, *Kronenberg et al*., 1984], but whether water-enhanced diffusion is limited to the mineral surface or extends into the bulk crystal lattice is not clear. Here we present experimentally determined Arrhenius parameters for reordering rates in optical calcite heated under wet high pressure (100 MPa) conditions. We observe marginal increases in reordering rates under these conditions relative to rates observed for the same material reacted under dry, low pressure conditions. This contrasts with the orders of magnitude increase in oxygen diffusivity at the mineral surface when water is present, suggesting the latter effect arises from surface reactions that have minimal influence on the diffusivity of C or O in the bulk mineral. Our results imply that previously published reordering kinetics determined under dry, low pressure experimental conditions are generally applicable to geological samples that have been heated in the presence of water.

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EGGHSHELL CLUMPED ISOTOPE TEMPERATURES: IMPLICATIONS FOR DINOSAUR THERMOREGULATION

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Carbonate clumped-isotope (Δ_{47}) paleothermometry has been used to estimate the body temperatures of extinct dinosaurs [*Eagle et al*., 2011], giving results within the range of modern endotherms. However, only a single clade, the sauropods, was investigated and it is unclear whether this is due to endothermy or their size (i.e. gigantothermy). The evolutionary timing of endothermy within Dinosauria is still debated [*Grady et al*., 2014)], therefore we use Δ_{47} paleothermometry of well-preserved dinosaur eggshells to test if smaller dinosaurs also had warm body temperatures and if there is a difference in body temperatures between dinosaurs more closely (i.e. *Troodon)* and more distantly (i.e. *Maiasaura*) related to birds [*Zelenitsky et al*., 2002].

Well preserved (based on SEM and trace elementmetal analysis) *Troodon* and *Maiasaura* eggshells from one locality give Δ_{47} temperatures within the range of modern endotherms (~ 37-42 °C). Poorly preserved hadrosaur eggshell from the same locality has a Δ_{47} temperature (~ 32 °C) closer to the Δ_{47} estimated paleoenvironmental temperature (~ 28 °C; based on a well preserved bivalve). A well preserved *Troodon* eggshell from another locality gives a Δ_{47} temperature ~ 27 °C, similar to a poorly preserved hadrosaur eggshell and a well preserved freshwater gastropod (~ 28 °C and ~ 26 °C, respectively) from the same locality. If our preservation interpretations are correct, dinosaur metabolism of those closely related to birds allowed for a range in body temperatures of at least ~ 10 °C. The variability in *Troodon* Δ_{47} estimated body temperatures may be due to different behavioral habits, as in modern ectotherms, or indicative of mesothermic metabolisms as in modern sharks and tuna [*Grady et al*., 2014].

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New developments in high-resolution gas source isotope ratio mass spectrometers

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Gas source isotope ratio mass spectrometry (IRMS) is one of the main tools for the study of the isotopic compositions of light elements, extended in the last 10 years to the measurements of molecules bearing several rare isotopes (e.g., clumped isotopes of CO_2) as well as position-specific isotopic substitutions in a few choice analytes (e.g., in N_2O). Measuring those low-abundance species creates several technical challenges, with the main one being the presence of numerous isobaric interferences. Those can come either from contaminants (background gases present in the source of the instrument or impurities introduced with the analyte), or unwanted beams created by the analyte itself during the ionization process (for example adducts and fragments).

In order to avoid those isobaric species, new high-resolution, double-focusing IRMS have been developed. We present here the capabilities of the production series version of the ThermoFisher Scientific 253 Ultra, which was installed at SUERC in July 2015. The instrument is capable of reaching high mass resolving power (above 40,000).

The instrument is similar in design to the Caltech 253 Ultra prototype. The collector array has 9 detector positions, 8 of which are movable. Faraday cups at each detector can be linked to amplifiers with gains ranging from 3.10^8 to 10^{12} Ohm (and 10^{13} Ohm amplifiers being currently developed). There are also 4 ion counters, one of which located behind a retardation lens (RPQ) to limit background noise and improve abundance sensitivity.

Additionally, one of the Faraday cup in the new instrument has a very narrow entrance slit, allowing high mass resolving power and high resolution, with a complete separation of the ion beams instead ofcomplex peak shapes corresponding to overlapping ion beams. This will potentially remove the need for adduct lines or peak stripping schemes for analytes like CH_4.

The temperature dependence of Δ_{47} thermometers and the exchange of CO_2 oxygen in phosphoric acid baths

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As the number of thermometric Δ_{47} calibrations grows, the evidence has strengthened that the slope of the Δ_{47} vs. inverse temperature relationship clusters into two distinct groupings based on whether the carbonate minerals were digested at low temperature (25°C) vs. high temperature (generally 70 - 90°C). We show that small amounts of oxygen exchange between CO_2 liberated to phosphoric acid solution and trace water in that solution can result in modification of the Δ_{47} signal. This can occur with very little shift in δ_{47} and $\delta^{18}O$ of the CO_2. We model this effect and provide initial experimental Δ_{47} measurements conducted using phosphoric acid produced with different $\delta^{18}O$ compositions. The extent and effect of oxygen exchange following $CaCO_3$ dissolution is a complex function of the phosphoric acid's temperature, water content, and viscosity. Conventional approaches to preparation of phosphoric acid (targeting a defined density range at room temperature) are likely inadequate for resolving the finer details of these dependencies. This stems in part from changes that can occur to acid water content and density during heating of the acid under vacuum. Secondary effects may also arise that relate to the grain and crystallite size of the carbonate minerals, intercalation of carbonate minerals with siliciclastic matrix in sediments or paleosols, and possibly the reactivity of hydrous components of a siliciclastic rich sample.

Biological signals in clumped isotopes of brachiopod shells

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Clumped isotopes offer the potential to unlock the rich environmental and biological record within marine carbonate biominerals. The co-occurrence, or 'clumping', at thermodynamic equilibrium, of two heavy isotopes, such as ^{13}C and ^{18}O, in carbonate molecules, records only temperature [e.g. *Eiler*, 2007; *Eiler*, 2011; *Schauble et al*., 2006]. Brachiopods have a long continuous fossil record of stable low-Mg calcite shells [*Parkinson et al*., 2005]. Since brachiopod primary (outer) and secondary (inner) shell layers form at the same temperature they would be predicted to have the same degree of clumping. We show that in examples of two brachiopod species, *Neothyris lenticularis* and *Laqueus rubellus*, there are biological signals in the clumped isotopes.

Temperatures calculated from measured $\delta^{18}O$ and known $\delta^{18}O$ of the seawater from which the brachiopods were collected, using the equation of Parkinson et al. [2005], are within the measured temperature range of the seawater from which they were collected. Temperatures calculated from clumped isotope Δ_{47} values, using the calibration of Henkes et al. [2013], are well above the measured temperature of the seawater from which they were collected. These higher temperatures reflect higher Δ_{47} values and therefore a lesser degree of clumping.

Primary and, particularly, secondary layers are "anticlumped" with less clumping than expected. Biomineral formation is controlled by enzymes that may prevent the attainment of thermodynamic equilibrium and thus influence the extent of clumping. Understanding these biological signatures in clumped isotopes, and determining the extent to which they are widespread among biominerals, offers an incisive tool to investigate biomineralisation.

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Uncertainties and standardization in the absolute reference frame

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Mass spectrometric methods for clumped isotopes have evolved considerably over the past decade, in large part because of the need to correct for various “nonlinearities”, which may be negligible in conventional analysis of singly-substituted isotopologues but manifest as large bias in clumped isotope measurements. In the case of CO_2 and carbonates, correcting for these nonlinearities yields Δ_{47} values expressed in the "absolute" reference frame of *Dennis et al*. (2011). There is, however, no precise consensus regarding the actual implementation of these various corrections, including the use of in-lab and international standards.

We propose that a useful way forward is to quantitatively assess the statistical uncertainties associated with the various corrections we all perform in one form or another. Comparing the merits and limitations of various standardization strategies illustrates how the optimal one will vary between laboratories, and highlights the critical need for one to three reliable international carbonate standards. Although several of these strategies are already used, more or less formally, we believe that this comparison exercise provides a good starting point for discussion.

Dennis et al. (2011) Defining an absolute reference frame for 'clumped' isotope studies of CO2. *Geochim. Cosmochim. Acta* 75, 7117– 7131.

High-precision measurements of isotopic anomalies in CO_2 using a new type of cavity ring-down spectrometer

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The application of laser absorption techniques to the measurement of relative isotopic abundances in gases has been making rapid progress over the past decade. Compared to mass spectrometry, these non-destructive methods offer specific advantages such as the ability to discriminate between isobaric isotopologues (e.g., $^{16}O^{13}C^{16}O$ vs $^{16}O^{12}C^{17}O$) or contaminants. Their precision, however, still lags behind that of modern mass spectrometers for several, mostly technical reasons.

Here we report on the development and early operational results from a new type of ultra-precise cavity ring-down spectrometer based on the combination of several novel techniques. The design goal of this OFFS-CRDS ("optical feedback frequency-stabilized cavity ring-down spectrometer") is to measure ^{17}O anomalies in CO_2 with a precision of 10 ppm, and a natural extension of this work is to aim for $^{18}O^{13}C^{16}O$ measurements with a similar precision.

A new design for automated CO_2 extraction and cleaning for $\Delta^{47}C$ measurements

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Measurements of clumped isotopes are time consuming and generally require the continual, or at least frequent, presence of an operator. To reduce required person-hours while maximizing information gathered during sample processing, we designed an automated extraction line that maximizes the time between human interventions while delivering a clean, unfractionated CO_2 separate. This line can be configured either as an online version (attached to the mass spectrometer) or as a sample port version that traps gas for later measurement. The major departure from other common acid bath autolines that rely on automated lifters and large open vessels of various fluids is a new design of the gas traps and chromatographic column that rely on heat exchange from closed loop pumped fluids of varying temperatures.

The slush traps and cryogenic focus traps were designed to minimize the number of moving parts, improve efficiency of consumables such as helium and liquid nitrogen and reduce human input. These traps are based around the principle of heat exchange, where pipes and a heater wire are nested together. In both trap types, a 1" diameter copper pipe forms the backbone, with a 1/8" diameter 316 stainless steel tube wrapped around this backbone. In the cryogenic focus traps, a ¼" copper tube is also wrapped around this backbone, forming a nested helix with the stainless steel tubing; heater wire is then wrapped around this helix. These elements are all coated with a highly thermally conductive epoxy (Duralco 132P) to insure they act as a single thermal mass. The sample will traverse the 1/8" 316 stainless steel tube and thus be subject to the temperature of the thermal block; temperatures can be changed by forcing a fluid through one of the other tubes or the main pipe, or by increasing the temperature of the heater wire. We plan to use the backbone pipe to pump ca. -80°C ethanol, and the ¼" copper tube to suck liquid nitrogen (similar to the VG SIRA and OPTIMA cold fingers). This should allow the thermal block to reach liquid nitrogen temperatures when the vacuum pump is pulling; these temperatures will be ca. -80°C when the ethanol is flowing, and above 20°C via the heater wire. To further clean samples for clumped isotope analysis, we will use a 2-stage Peltier cooler to chill a small, insulated box (similar to a GC oven) containing a column packed with Poropak. The box will be heated with a resistance heater when the chromatographic column is not in use.

Computer control and automation will be attained with an Arduino Mega microcontroller board. This controller will also be used to log temperature, vacuum pressure, sample yield, and gas composition data (from an attached residual gas analyzer) for all samples. The controller will also monitor for problems, such as insufficient sample size, and take appropriate action. If problems requiring human intervention are encountered, such as a vacuum leak or insufficient liquid nitrogen in the system, the controller can notify the appropriate person.

Analytical effects on clumped isotope thermometry: Comparison of a common sample set run using multiple instruments and methods

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Clumped isotope geochemistry requires extremely high analytical precisions compared to most traditional single-isotope measurements, as the target isotopologues are extremely rare. Measurements are complicated by several analytical challenges, including variable sample digestion and clean up procedures, molecular fragmentation and recombination in the source [*Huntington et al*., 2009], the generation of interfering ions, pressure-dependent baselines [*He et al*., 2012; *Bernasconi et al*., 2013], and drift in equilibrated gas slopes and scale compression factors [*Meckler et al*., 2014]. An outstanding question is whether differences in the application of these corrections and other operating factors may contribute to measurement discrepancies between lab groups. Here we explore these issues through the analysis of a suite of synthetic carbonates grown at controlled temperatures as well as internal laboratory standards using measurements on two different mass spectrometers, and compare them to a previously published dataset of the same materials [*Defliese et al*., 2015]. The materials are analyzed using different extraction and purification techniques, and measured on both a Thermo Fisher MAT 253 and a Nu Instruments Perspective mass spectrometer. We compare the effects of different background correction and standardization techniques on the same dataset, such as standardization to the absolute reference frame [*Dennis et al*., 2011], including using only equilibrated gases versus a mixture of gases and carbonate standards, and carbonate standards alone.

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Microbially-mediated and abiotic synthesis of siderite from 10 to 70°C: a new Δ_{47} and $\delta^{18}O$ calibration

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The clumped and oxygen isotope composition of pedogenic siderite ($FeCO_3$) can be exploited to reconstruct climate in humid continental environments. The only published siderite Δ_{47}-T calibration [Fernandez et al. 2014] is based on few data points and does not cover the entire temperature range that can be expected in palaeosols. Furthermore, interlaboratory discrepancies regarding Δ_{47}-T calibrations are yet to be resolved. For oxygen isotopes, two calibrations exist. One is derived from abiotic [Carothers et al. 1988] and the other from microbially-mediated precipitation experiments [Zhang et al. 2001]. These calibration are are consistent between 45 and 75°C, but but diverge below 33°C and disequilibrium between DIC and water cannot be ruled out [Zhang et al. 2001]. In this contribution, we present a new siderite Δ_{47} and $\delta_{18}O$ calibration that was produced by inorganic precipitation through active degassing and with microbial cultures using the iron reducing bacteria *Shewanella Putrefaciens*. Our aim is to cover a temperature range from 10 to 70°C. To promote DIC equilibrium, the enzyme carbonic anhydrase is used in both active degassing [Fernandez et al. 2014] and bacterial culture experiments. Experiments between 25 and 35°C will be performed both inorganically and with bacterial cultures. Siderites were digested offline at 100°C and analyzed with a MAT 253+ isotope ratio mass spectrometer (Thermo Scientific, Bremen, Germany). Δ_{47} values were pressure baseline corrected, transferred to the absolute reference frame and projected on an acid digestion temperature of 25°C [Meckler et al. 2014]. Preliminary results based on 4 precipitation temperatures indicate that the derived Δ_{47}-T calibration has the same temperature sensitivity as the previously published calibration of Fernandez et al. [2014]. Furthermore, both Δ_{47} and $\delta^{18}O$ to temperature regressions are highly linear. Δ_{47} and $\delta^{18}O$ fractionation at 30°C is the same for both abiotic and microbially mediated siderites.

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Δ_{47} clumped-isotope analyses at the University of Wollongong

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The current clumped-isotope configuration at the University of Wollongong utilises side-arm acid reaction vessels with an acid reaction temperature of 25°C in order to directly compare results to the conventional 25°C reaction temperature of the original clumped-isotope studies and subsequent calibrations. Purification is achieved by cryogenic trapping of water vapor, evacuation of non-condensable gases followed by gas chromatographic purification. We employ an Agilent® 7890A GC and HP-PLOT-Q fused silica polystyrene-divinylbenzene column (30m, 530µm ID) held at -20°C, with a carrier gas helium flow of 2ml/min. The CO_2 signal is monitored for contamination through a non-destructive thermal conductivity detector (TCD). This system is off-line in relation to both the extraction line and mass spectrometer.

Measurements are made on a ThermoFisher MAT253 using an acceleration potential (HV) of approx. 9.5kV with the "sulfur-window" closed. Each sample measurement consisted of 12 acquisitions of 10 cycles, each cycle comprises one set of 26s integrations of reference gas and sample gas through the change-over valve. Between each individual acquisition an automatic bellows pressure balance was performed to yield 8V on mass 44 on both the sample and reference gas bellows. A background measurement was performed before each acquisition set. Peak centering is performed manually on cup 2 at the beginning of a measurement set and is rechecked at the end of a measuring set. If the peak center has changed substantially or was centering off-peak, the measurement set is discarded. Individual cycles have a duration of approximately 20 to 30 minutes. This equates to an approximate analysis time of 4 hours per sample. The capillary flow rate is adjusted to yield a 2V signal on mass 44 at a bellows pressure of 20mbar. Capillary balancing is checked weekly with a maximum capillary misbalance of 0.5% between sample and reference capillary flowrate.

Δ_{47} values are expressed in the absolute reference frame constructed using heated-gases (Oztech reference gases), water-equilibrated samples at 10°C, 25°C and 50°C (± 0.1°C) using isotopically different equilibration waters (A-1; Antarctic melt-water, C-1; Canberra rain water and P-1; Perth rain-water). The resulting empirical transfer function (ETF) of $\Delta_{47\text{-RF}} = 1.000580421\ \Delta_{47\text{-[EGvsWG]0}} + 0.839633943$ with an $r^2 = 0.9987$ was used to transpose all measurements into the absolute reference frame. Reproducibility of carbonate CO_2 analysis was found to be 0.02, 0.05 and 0.009‰ (1σ) for $\delta^{13}C$, $\delta^{18}O$ and Δ_{47} respectively, determined through analysis of Carrara Marble (CM-1), ETH standards and an in house analytical-grade calcium carbonate (AR). The Carrara Marble standard produced Δ_{47} values of 0.434‰ (SE 0.005) corresponding to a temperature of 121.7±2.7°C.

The Orbitrap as a gateway to high-dimensionality studies of molecular isotopic structure

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A promise of clumped and position-specific isotope geochemistry is that they may grow to encompass the vast isotopic diversity of molecules, providing many constraints on their origins and histories. The reality falls short of this vision: even the most technologically advanced proven tools examine only a handful of isotopologues. We will present the instruments, methods, initial results and strategic goals of an experiment that aims to advance this field past a key threshold we term the '100 isotopologue challenge': Analysis of 100 isotopologues of a single compound at ~1 ‰ precision, on ~micro-molar samples, in a day.

Our study focuses on several parallel problems: Developing a technology capable of constraining dozens of singly- and multiply-substituted forms of an analyte with the requisite sensitivity, precision and speed; processing and interpreting those constraints into statements of proportions of isotopologues; definition of a reference frame of theoretical predictions and experimental observations that will let us interpret data for samples; and establishing ways of summarizing and depicting measurements of such complexity. Success will require the convergence of advances at all of these technical and conceptual problems, and aiming the resulting capability to worthy applications.

We focus our energies on Fourier transform mass spectrometry (FTMS), which has several properties that could be transformative for our goals, including: exceptional mass resolution (M/ΔM of hundreds of thousands); the ability to simultaneously observe many species spanning significant ranges in mass; and high sensitivity. The key question is whether this tool can be adapted into an analytical instrument with useful precision and accuracy in isotope ratios.

Initial experiments were performed using the Q-Exactive GC— a novel version of the Orbitrap family of FTMS's having a gas ion source, GC sample introduction system, and optics optimized for study of relatively low mass ions. Measurements of D/^{13}C of isobutene demonstrate the capacity of this instrument to return isotope ratios near natural abundances, with excellent ratio stability (varying little over many hours), and external errors down to 1 per mil. This instrument enables a high level of operator control on the portions of the mass spectrum interrogated in each measurement, and thus should be suitable for quickly assembling data sets constraining the isotopic contents of molecules and several of their fragment ions. Prior studies demonstrate that this instrument has minimum detection limits on the order of a few ions (in a ~1 second time window), meaning it should be possible to extend these capabilities to relatively rare, multiply substituted species. In the coming months we will explore the limits of sensitivity and precision for this platform and design the first methods that retrieve proportions of ~10 isotopologues from the mass spectra of simple model compounds — our beach head on the 100 isotopologue challenge.

Clumped isotope analyses of large benthic foraminifera

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The oxygen isotope and Mg/Ca ratio of foraminifera tests relate systematically to temperature, and have been widely and successfully applied within the recent geological past. However, both proxies are dependent on the respective ratio in seawater, which significantly complicates the interpretation of such datasets. Foraminifera have been successfully utilised for clumped isotope analysis [Grauel *et al*., 2013; Tripati *et al*., 2010], yet the small mass of most species precludes their widespread use, especially given the simultaneous requirement for exceptionally-preserved material. We present clumped isotope data for a group of large benthic foraminifera species, which are always associated with photosymbionts in the modern ocean and are thus restricted to the upper 100 m of the water column in shelf and reef environments. This group of benthic foraminifera are well suited for clumped isotope analysis as their large size means that fewer than 10 individuals are required for a typical measurement. Previous work has shown that these organisms have promising utility for ocean temperature and seawater chemistry reconstruction during the Eocene [Evans *et al*., 2013], a geological epoch spanning 56 to 34 million years before present which was characterized by atmospheric CO_2 concentrations several times greater than modern. During this time, these large foraminifera were globally distributed within the low-mid latitude oceans, and abundant to the extent that they are the main component of some shallow water carbonates.

Our data show that live-collected large benthic foraminifera of the genus *Operculina* precipitate calcite with a clumped isotope signature in tight agreement with the inorganic calcite data of Zaarur et al. [2013]. This allows us to derive near-surface ocean palaeotemperatures by applying this calibration to fossil specimens. We present data from 10 globally-distributed (sub)tropical sites, with an aim to more accurately constrain the magnitude and latitudinal extent of Eocene warmth. Preliminary results indicate that the western Pacific was substantially warmer (~6°C) during the Eocene, whilst the mid latitudes (southern UK) was characterised by mean annual sea surface temperatures more than twice present-day.

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Small Changes of Paleoclimate and Paleoenvironment across the Eocene–Oligocene Transition in the Central Rocky Mountains

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Earth's climate experienced dramatic cooling during the Eocene–Oligocene Transition (EOT) based on marine oxygen isotope records. How climate and environment in continental interior responded to this cooling event are not well understood. Here we study clumped isotope temperatures and oxygen isotope compositions of carbonate cements and bulk organic carbon isotope compositions in order to reconstruct paleoclimate and plaeoenvironment across the EOT in the central Rocky Mountains, western U.S.A. The White River Formation in the eastern Wyoming is a well-dated stratigraphic unit that contains tuffaceous mudrock and sandstone deposited in fluvial and eolian depositional environments. Because the carbonate cements in the formation were formed as low-Mg calcite in equilibrium with fresh surface water during early diagenesis, the isotope compositions reflect near-surface climatic and environmental conditions. Our preliminary results of Δ_{47} vary from 0.682 ‰ to 0.713 ‰ in the Absolute Reference Frame, corresponding to carbonate crystallization temperature of 35±2 ºC to 19±3 ºC using the composite calibration for CO_2 extraction at 75-100 ºC in Defliese et al. [2015]. The temperatures do not display any obvious trend suggesting no major temperature change in the central Rockies across the EOT. The carbonate $\delta^{18}O$ values increase from ~-11.8 ‰ before the EOT to ~11.3 ‰ after the EOT, which may reflect gradual drying or a 2 ºC decrease of temperature. The organic carbon $\delta^{13}C$ values remain stable across the EOT. Our results show that the paleoclimate and paleoenvironment in the central Rocky Mountains only experienced small changes across the EOT.

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Early Eocene latitudinal temperature gradient estimated from siderite clumped isotope thermometry

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The Eocene (56-34 My) is the youngest period of earth's history when CO_2 concentrations in the atmosphere (600-1500 ppm) reached levels close to those predicted for future emission scenarios. Paleoclimate data from this interval can, therefore, be used to understand the effects that anthropogenic emissions will have on the climate system. Here, we present clumped and oxygen isotope measurements of siderite samples collected along a North-South transect in the North American Continent. These siderites formed in soils under water-saturated conditions and provide a record of both soil temperature and the $\delta^{18}O$ composition of meteoric waters, which can be used to unravel paleoprecipitation amounts. Soil temperatures and meteoric water $\delta^{18}O$ values were estimated using an internally derived calibration constructed with synthetic siderite samples precipitated in the presence or absence of iron reducing bacteria. Our data provide an estimate of the latitudinal temperature gradient during the early Eocene and offer a test dataset for models of the earth's hydrologic cycle during greenhouse conditions.

A numerical framework for interpreting clumped isotope 'vital effects' in scleractinian coral skeletons

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The carbonate clumped isotope thermometer constrains carbonate formation temperature based on the extent of ^{13}C and ^{18}O clumping in the carbonate mineral lattice (Δ_{47}), and is a promising tool in paleoclimate and paleoceanographic research. However, there is growing evidence that biological processes ('vital effects') can cause deviations in the clumped isotope compositions of scleractinian coral skeletons from the expected thermodynamic equilibrium values in certain coral genera. This finding complicates the application of this thermometer in corals.

Based on existing models of stable isotope vital effects in corals, we present a numerical framework for interpreting clumped isotope vital effects in scleractinian coral skeletons. This framework takes into account both the kinetic isotope effects associated with CO_2 hydration/hydroxylation reactions and isotope equilibration of dissolved inorganic carbon (DIC) in the coral's calcifying fluid. It allows quantitative evaluation of the effects of a variety of factors on the oxygen isotope and clumped isotope vital effects in corals. These factors include temperature, pH of the calcifying fluid, the source of carbon for calcification, concentration of carbonic anhydrase and residence time of DIC in the calcifying fluid. Our model results suggest the source of carbon for calcification (i.e. CO_2 diffused across cell membrane versus DIC from seawater leakage) and the isotope exchange kinetics of DIC in the calcifying fluid place the first-order control on the magnitude of clumped isotope vital effects present in coral skeletons. A higher fraction of diffused CO_2 as the carbon source and a smaller extent of isotope exchange within the calcifying fluid are expected to lead to larger clumped isotope vital effects. Our model reproduces the different types of relations observed between clumped isotope and oxygen isotope vital effects in different coral genera, and supports the idea that genus- or species- specific calibrations could be a promising approach for future application of the clumped isotope thermometer in coral skeletons.

The accuracy of our model predictions is limited by the lack of constraints on the values of model parameters. To enable better comparison between model results and experimental observations, future studies should aim to place better experimental constraints on these model parameters in different coral genera.

Clumped isotope thermometry of distal deposits from the end-Cretaceous Chicxulub impact event

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The petrography and geochemistry of carbonate accretionary particles (lapilli) from Cretaceous-Paleogene (K-Pg) boundary sediments exposed along the Brazos River, TX suggest a unique calcite formation environment in the ejecta blanket immediately following the Chicxulub impact event (Yancey, 1996; Yancey and Guillemette, 2009). Clumped isotope thermometry applied to these small lapilli (~0.05-0.3 cm) can help further constrain the conditions and timing of their formation. Our initial hypothesis was that these calcites formed relatively quickly during the expansion and subsequent cooling of the extremely hot (>1000 °C) post-impact ejecta cloud. Because the kinetics of carbonate clumped isotope bond ordering are predicted to be nearly instantaneous at such high temperatures, we expected clumped isotope temperatures ($T(\Delta_{47})$) from quickly cooled lapilli to reflect similarly high temperatures at or near the stochastic prediction of ^{13}C-^{18}O bond ordering (Schauble et al., 2006; Guo et al., 2009). Our observations, however, suggested a different formation environment. Lapilli $T(\Delta_{47})$ range from 60-230 °C despite little variability in measured carbonate $\delta^{18}O$ (-6.2 ± 1.1‰), which points to lower temperature closed-system processes dominating the clumped isotope signal. We believe this to be consistent with calcite formation via portlandite ($Ca(OH)_2$) carbonation reactions in cooler post-impact atmospheric conditions that are consistent with this mineral's stability field (Yancey and Guillemette, 2009), though other formation mechanisms are certainly plausible. Indeed, correlations between $\delta^{13}C$, $\delta^{18}O$, and $T(\Delta_{47})$ indicate 'bulk' isotopic evolution of lapilli with during apparent cooling. Finally, to partially rule out thermal and isotopic diagenetic overprinting of lapilli, we measured benthic foraminifera and a mudstone from the same locality. Earth surface-like $T(\Delta_{47})$ from these samples rules out the hypothesis that the Brazos River section carbonates were effected by clumped isotope bond reordering during deep sedimentary burial, which is supported by independent burial history estimates. Benthic foraminifera clumped isotopes may also be a new window into the ancient Gulf of Mexico shelf paleoenvironments immediately before and after this important geological boundary and event, though more replication of emerging trends is required.

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Latest Developments in High Resolution Isotope Ratio MS

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Thermo Fisher Scientific uses 10 kV accelerating voltage for the highest performance isotope ratio mass spectrometers, currently MAT 253, 253 Plus and 253 Ultra (IRMS), Triton plus TIMS, Neptune plus MC ICPMS, and Helix MC and SFT static source mass spectrometry. Thermo uses 10 kV for applications which require the highest sensitivity; best peak shape, peak stability, and abundance sensitivity; and, most recently, achieving high mass resolution, which attributes are important for very small isotope signatures related with small isotope abundances and extended dynamic ranges (analysis of isotopomers and of isotopologues containing clumped isotopes signatures). Because the quality of ion beams at such low abundances can be affected by interfering isotopologues and isobaric ions of contaminants and adducts with the same nominal mass, the method of choice for accurate quantification of low abundant ion beams involves resolving such interferences by medium or high mass resolution.

The new 253 ULTRA high resolution double focusing 10 kV IRMS has extensive multicollector capabilities, including the new 10^{13} Ω technology, which allows the analysis of isotope clumping in intact molecules. The understanding of clumped isotope effects is allowing direct position specific isotope analysis and is leading to a completely new field of isotope ratio analysis, isotomics. We will present an update on our latest improvements and recent developments on the 253 Ultra-high-resolution double focusing IRMS. Recent applications performed on the 253 Ultra will be presented.

Effect of photosynthesis on the abundance of $^{18}O^{13}C^{16}O$ in atmospheric CO_2

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The abundance of the isotopologue $^{18}O^{13}C^{16}O$ (Δ_{47}) in atmospheric air is a promising new tracer for the atmospheric carbon cycle (Eiler and Schauble, 2004; Affek and Eiler, 2006; Affek et al., 2007). The large gross fluxes in CO_2 between the atmosphere and biosphere are supposed to play a major role in controlling its abundance. Eiler and Schauble (2004) set up a box model describing the effect of air-leaf interaction on the abundance of $^{18}O^{13}C^{16}O$ in atmospheric air. The main assumption is that the exchange between CO_2 and water within the mesophyll cells will imprint a Δ_{47} value on the back-diffusing CO_2 that reflects the leaf temperature. Additionally, kinetic effects due to CO_2 diffusion into and out of the stomata are thought to play a role. We investigated the effect of photosynthesis on the residual CO_2 under controlled conditions using a leaf chamber set-up to quantitatively test the model assumptions suggested by Eiler and Schauble (2004).

We studied the effect of photosynthesis on the residual CO_2 using two C_3 and one C_4 plant species: (i) sunflower (*Helianthus annuus*), a C_3 species with a high leaf conductance for CO_2 diffusion, (ii) ivy (*Hedera hibernica*), a C_3 species with a low conductance, and (iii), maize (*Zea mays*), a species with the C_4 photosynthetic pathway. We also investigated the effect of different light intensities (photosynthetic photon flux density of 200, 700 and 1800 μmol $m^{-2}s^{-1}$), and thus, photosynthetic rate in sunflower and maize.

A leaf was mounted in a cuvette with a transparent window and an adjustable light source. The air inside was thoroughly mixed, making the composition of the outgoing air equal to the air inside. A gas-mixing unit was attached at the entrance of the cuvette that mixed air with a high concentration of scrambled CO_2 with a Δ_{47} value of 0 to 0.1‰, with CO_2 free air to set the CO_2 concentration of ingoing air at 500 ppm. The flow rate through the cuvette was adjusted to the photosynthetic activity of the leaf so that the CO_2 concentration at the outlet was 400 ppm and varied between 0.6 and 1.5 L min^{-1}. CO_2 and H_2O concentrations in air were monitored with an IRGA and air was sampled at the outlet with flasks.

We found that the effect on Δ_{47} of the residual CO_2 for the C_3 species sunflower and ivy was proportional to the effect on $\delta^{18}O$ of the residual CO_2. The difference in Δ_{47} between the in- and outgoing CO_2 was between 0.13 and 0.47‰, varying with the CO_2 concentration in the chloroplasts relative to the bulk air (C_c/C_a). The C_c/C_a depends on conductance and photosynthetic activity, and was different for the two species and was manipulated with the light intensity. No effect on Δ_{47} was observed for the C_4 species maize. This may be related to its lower C_c/C_a ratio and possibly a lower carbonic anhydrase activity causing incomplete exchange with leaf water. We will discuss these results in light of the suggested fractionation processes and discuss the implication for the global Δ_{47} value of atmospheric CO_2.

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Calibration Mixtures to Improve Accuracy of IRMS Measurements

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The dependence of stable isotopic analyses in geochemistry and environmental measurements are steadily increasing. However, the ability to compare measurements between laboratories can be quite challenging owing to differences in measurement and calibration. The key to obtaining reliable data is by designing experiments which utilize sampling methodologies that represent the environment intended for the study and meet the data-quality objectives. These samples in turn must then be calibrated against suitable reference materials containing low levels of uncertainty. The precision and accuracy of the analytical result is directly related to the precision and accuracy of the standard used to calibrate the analytical instrument.

The preparation of high quality calibration mixtures which can be used as analytical standards typically requires the use of cylinders manufactured from aluminum alloys. The exact choice of alloy may be governed by local authorities. The cylinder valve selected must also consider material compatibility of the various components in the mixture to ensure stability of the mixture. Traces of moisture and oxygen contained within the cylinder are known to cause stability issues in mixtures and pure gases. Metal surfaces can adsorb water that is tightly bound to the surface. Therefore the use of suitable vacuum baking procedures must be followed. Calibration gases can either be prepared gravimetrically, volumetrically or by dynamic blending. Each of these methods has its own advantages and limitations. Gravimetric techniques have the advantage of high mass discrimination and the use of weights which are directly traceable to national metrology institutes. One of the most significant limitations o f this technique is that the balance used must have a large range and suitable sensitivity for the addition of the minor components. Gravimetric mixtures are prepared by accurately measuring gas additions as measured by the increase in mass of the cylinder. The composition of the mixture is calculated based on the mass of each gas as well as some additional parameters. Chief among them is the purity of the source materials used. The individual source materials must be carefully characterized to take impurity constituents (both molecular and isotopic) into account in the final calculation.

Errors and uncertainties can arise from three primary sources (1) analyzer calibration (2) analyzer repeatability (precision) and (3) uncertainties in concentration of reference materials used for calibration.[1] The contribution to the uncertainty can be obtained by propagating the standard uncertainties of each source. To decrease the uncertainty, each parameter must be minimized. This entails calibrating the instruments with the highest accuracy standards.

Solid Isotopic primary reference materials, which can be combusted, are available in extremely limited quantities from the International Atomic Energy Agency (IAEA) and the US National Institute of Standards and Technology (NIST), formerly known as NBS. However, quantities of these materials are typically limited to one reference material per laboratory every 3 years. Laboratories are encouraged to develop their own working standards that can be used to calibrate samples. Preparation of these standards often increases the total uncertainty of the measurements. It is good analytical practice to matrix match the standard and sample as well as bracket the standards around the concentration of the samples. Typically, 3 standards are the minimum required to assess linearity and accuracy. How can this be accomplished with the limited range of standards available?

This presentation will examine the isotopic mixture preparation process, for both molecular and isotopic concentrations, for a range of components and delta values. The role of precisely characterized source material will be presented. Analysis of individual cylinders within multiple batches will be presented to demonstrate the ability to dynamically fill multiple cylinders containing identical compositions without isotopic fractionation. Additional emphasis will focus on the ability to adjust isotope ratios to more closely bracket sample types without the reliance on combusting naturally occurring materials, thereby improving analytical accuracy.

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Improving and Simplifying Clumped Isotope Data Archiving, Corrections, and Evaluation using Easotope

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One of the major challenges facing clumped isotope expert is data treatment in a time efficient and consistent manner. This problem is compounded by the fact that clumped isotope data acquisition is very long, and that data needs to be corrected using a series of external standards acquired over a period of several weeks. Ensuring consistency of the correction between individuals within a research group becomes problematic, and inter-laboratory consistency even more so. A further complication is that the recent pressure baseline methods (PBL) developed for non-linearity correction (He et al 2012, Bernasconi et al 2013) require corrections of the raw voltage intensities, which is time consuming thus limiting the ability to freely test different background selections for PBL.

Here, we present new software ("Easotope") that addresses the problems faced by the clumped isotope community specifically, but also more generally by any community dealing with complex corrections for clumped isotope systems. Easotope is open-source, freely available software written in Java. The software is a full-fledged database that can store and read raw data files, can be run locally on a single computer, or distributed across a client-network architecture. Easotope is platform-independent, and at the time of writing this abstract runs as a "beta" version at Imperial College London (were it was conceived) and at the ETH Zurich. This practical demonstration will highlight the benefits of Easotope, which include a full archive and easy access to the raw data, ease and consistency of the corrections, transparency of data and references used in the corrections, graphical output, and the ability to explore alternative ways of correcting the data in a fraction of the time required with other softwares. Easotope handles heated gas corrections (Huntington et al 2009), projection into the absolute reference scale using both primary and secondary transfer functions (Dennis et al 2011), pressure baseline correction using the "Chicago" (He et al 2012) and "Zurich" methods (Bernasconi et al 2013), acid fractionation correction for both bulk and clumped isotope data, and temperature calculation.

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A Review of the Performance of the "Imperial Batch EXtraction" (IBEX) Device for Clumped Isotope Analysis

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Because clumped isotopes are based on measuring rare isotopologues, a long-integration time to increase signal-to-noise ratio and thorough cleaning of the CO_2 gas to avoid contaminants are important prerequisites. As a consequence, relatively large samples are required, and the total integration time on the mass spectrometer is long (>6-9 hours/sample). Furthermore, the cleaning procedure used to prevent contaminated gas to enter the mass spectrometer is intricate and time consuming. The long preparation time coupled with the need for data consistency favors the use of automated lines: these allow for data acquisition over prolonged periods of time, and avoid caveat in sample preparation that typically arise when researchers of varying experience use a manual vacuum system. The "Imperial Batch EXtraction" (IBEX) is the first fully-automated preparation device using gas chromatographic separation designed for clumped isotope analysis. Other semi-automated devices coupled to GC systems exist at CalTech and elsewhere, but these generally do not include automatic liquid nitrogen refill. Alternatively, the Thermo Kiel device can be modified to include a poropak trap, but this lacks a full-GC capability with a carrier gas and thus may not be capable of cleaning difficult samples.

The IBEX design includes a common acid bath set at the operator's temperature of choice, a multi positions (currently 40) auto-sampler, and a series of automated traps. The CO_2 resulting from carbonate acidification is first capture in a trap set at liquid nitrogen temperature, then released by heating the trap to -100˚C. The gas is passed using a carrier gas (helium) through two sulfur traps containing silver wool and a 1-meter GC column containing poropak-Q. The gas is then captured in a second water-CO_2 trap, and finally transferred to a micro-volume before being inlet into the mass spectrometer for clumped isotope analysis. The traps are cooled using liquid nitrogen stored in a central dewer that is automatically refilled from a tank when needed when needed. The IBEX has dedicated control software written in C-Sharp, a method editor allowing the operator to change the analysis parameter, and the system is currently fully integrated with the Thermo Isodat software. The IBEX also has the capacity to run on any IRMS currently available.

This presentation will focus on the results obtained on the IBEX, and the data will be shown using Easotope, a free software for clumped isotope corrections (John and Bowen, this meeting). We will discuss the reproducibility of the IBEX for both clumped isotopes and bulk isotopes, and show data from both standards and unknown samples. Of particular interests is the reliability of the IBEX, and how well samples can be cleaned using this system. Initial results suggest that sample cleaning is superior to the manual line.

Sample size is another important parameter to consider: current routine replicate size is 3 mg of carbonates, but initial testing suggests that the IBEX performs well with samples of 2 mg carbonate, and that this could potentially be further reduced. In conclusion, the IBEX has proven to produce reliable data from both samples and standards in a fully automated way, and the typical sample size needed for one replicate is the smallest currently performed in dual inlet mode using the bellows on a MAT 253.

Dating and clumped isotope-based temperature of a paleo-jacuzzi (Vértesszőlős Early Man site, Hungary)

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One of the most important Early Man site of Europe is located at Vértesszőlős (Hungary). Here, settlements of Paleolithic people and remains of *Homo heidelbergensis* were preserved in loose sediments and travertine pools. The site has been investigated from many aspects, but still there are open questions regarding its age and the paleoclimatic conditions during the travertine deposition. In this study U-Th dating and stable- and clumped isotope analyses on 7 travertine samples were performed representing different travertine- and culture layers of the Early Man site. Stable isotope and Δ_{47} data were measured with a Thermo Fisher Kiel IV device coupled to a Thermo Fisher MAT 253 mass spectrometer at ETH Zürich [*Schmid and Bernasconi*, 2010]. The U-Th dates were determined at the HISPEC (NTU, Taiwan). The clumped isotope thermometer was calibrated recently using recent travertine and tufa deposits [*Kele et al*., 2015] and here we apply this calibration for first time on ancient travertines. Our Δ_{47} data indicate that the temperature of the precipitating water has varied between 13 and 21 °C (during MIS 9 and MIS 8) and it was 17 °C during the precipitation of the travertine containing the occipital bone of *Homo heidelbergensis*. The earlier indirect estimations based on malaco faunas suggested for a temperature similar to the present conditions, however, the malaco-thermometer indicates only the conditions of the vegetation period (mid-temperature of July). Based on earlier studies the age of the site was placed to the „Inter-Mindel" period or to the Upper Biharian stage after the biostratigraphy; recent studies, however, questioned the validity of these identifications. Previous U-Th datings yielded various ages ranging from 202 kys to >350 kys. According to our U-Th analyses the age of the Vértesszőlős travertine at the occipital bone is about 315±72 kys. The age of the eastern continuation of the foot-mark surface is 310±30 kys and the age of the lower part of the lowermost culture layer is 328±28 kys. These results shed new light to the absolute chronology of the travertine complex and the Lower Palaeolithic site.

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Burial History of the Neoarchean Campbellrand-Malmani Carbonate Platform

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We report a study of carbonates from the Campbellrand-Malmani carbonate platform of the Neoarchean (2.68-2.50 Ga) Kaapvaal Craton, South Africa. This sequence preserves the least altered late Archean rocks known to exist, with maximum metamorphic temperatures that range from 110°C to 150°C and 130°C to 170°C based on greenalite-quartz-magnetite-hematite and greenalite-quartz-minnesotaite equilibrium, respectively, in the overlying Kuruman Iron Formation [*Miyano and Beukes*, 1984].

For this study, we analyzed nine outcrop specimens from multiple members of the Gamohaan Formation. The strata are comprised primarily of stromatolitic limestone with localized early diagenetic dolomitization. Hand specimens have been micro-sampled according to mineralogy and carbonate textures to interrogate the most homogeneous material for analysis. These textures include sparry and "herringbone" calcite as well as lamellar micrite and dolomite. Additionally, spatial variability was investigated by sampling three different drill cores that have been described previously by Klein and Beukes [1989]. The three cores (AD-5, WB-98, and DI-1) cover a distance of 212 km and are located approximately along strike. A single, characteristic crytalgal limestone layer has been targeted in the cores and outcrop to ensure the most valid spatial comparison possible.

In addition to clumped isotope analysis, sample investigation also included petrographic and electron microscopy as well as electron backscatter diffraction of thin sections. Apparent temperatures determined by clumped isotope thermometry range from 102°C to 178°C for sparry calcite, 89°C to 151°C for micrite and "Herringbone" calcite, and 104°C to 153°C for dolomite. The average values for the aforementioned textures/minerals are 143°C, 131°C, and 124°C (±21°C), respectively. Systematic differences are not observed for samples collected from different locations along strike or stratigraphically in the outcrop samples. Apparent clumped temperatures are in good agreement with burial temperatures estimated by metamorphic mineral assemblages. Clumped isotope thermometry has allowed us to go a step further and determine a likely diagentic and burial history of these samples based on geochemistry, mineralogy, petrology, and modeled isotope-exchange reactions by solid-state diffusion [*Stolper and Eiler*, 2015].

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Triple-isotopologue Analysis of N_2 as a Tracer of the Global Nitrogen Cycle

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The uncertainty of the global nitrogen budget highlights the importance of a better understanding of the fixed-nitrogen inventory. High-precision measurements of $^{14}N^{14}N$, $^{14}N^{15}N$ and $^{15}N^{15}N$ in N_2 can provide new constraints on this inventory. We report the first measurements of these species in tropospheric N_2 and in N_2 generated from denitrification and discuss the potential uses of N_2 "triple-isotopologues" to improve our understanding of the nitrogen cycle.

The N_2 triple-isotopologue distribution, expressed as Δ_{30}, describes enrichment in $^{15}N^{15}N$ of N_2 relative to that predicted by chance. It reflects isotopic ordering through N-N bond breaking and formation during nitrogen fixation and denitrification. The Δ_{30} parameter is analogous to $\delta^{15}N$ in that it corresponds to chemical transition among N-bearing compounds. However, unlike $\delta^{15}N$, Δ_{30} is insensitive to the total $^{15}N/^{14}N$ ratio in the reservoirs.

We present Δ_{30} values of N_2 analyzed on a high-resolution isotope ratio mass spectrometer (the Nu instruments *Panorama*). The analytical accuracy was verified with laboratory Knudsen diffusion experiments. The Δ_{30} values of N_2 from soils and pure cultures of denitrifiers vary between +16.9 and +18.9‰, which indicate large kinetic isotopic effects during denitrification and probably reflect the isotopic signatures of enzymes that catalyze N-N bond formation. Enrichment in $^{15}N^{15}N$ was also observed in tropospheric N_2 (Δ_{30} = +19.05 ± 0.12‰, 1σ), which likely inherits its value from the nitrogen lost from biosphere. Overall, our results suggest that the degree of isotopic ordering of tropospheric N_2 may be applied to reflect the relative contributions of global natural N_2 sources.

Keywords: global nitrogen cycle, N_2 triple isotopologues, denitrification

Clumping Modern Speleothems - Correcting for Variations Seen in Calcite Δ_{47} Precipitated Under Known Conditions

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Clumped isotopic measurements of CO_2 have great potential in measuring paleotemperatures recorded in rocks and sediment without knowledge of the initial isotopic composition of the water. However, applications to speleothems have proven to be complex and difficult, requiring better methods for processing and calibration in order to accurately interpret temperatures measured in the calcite. Monitoring experiments have been set up in a cave with modern forming stalagmites on the island of Eleuthera in the Bahamas. Glass plates (curved and flat) have been placed underneath dripping stalactites for periods of 4-5 months to precipitate calcite under known conditions. Ambient cave temperature, relative humidity, air chemistry, and drip water chemistry have been monitored since the start of the project in 2012. Although the temperature logger left in the cave has reported an average temperature of 23.5°C over the duration of the experiment, clumped isotope Δ_{47} values have reported temperatures consistently 10-15°C warmer. Our study of modern cave conditions allows for opportunities to experiment with methods of data correction/calibration in order to be able to better interpret clumped isotopic temperatures measured in ancient stalagmites. Plotting systematic offsets from expected values of Δ_{47} and $\delta^{18}O$ measured in calcite samples from the modern cave gives a slope for disequilibrium. We are able to correct to the equilibrium line established by Tremaine et al., 2011 by using the disequilibrium slope from the modern. Although this correction does bring measured temperature values in the modern calcites close to the actual cave temperature, more work needs to be done to better this method for more universal application to all speleothems.

Late Cretaceous Sea Surface Temperature Variability of the Southeast U.S.

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The Maastrichtian (66 to 72 Ma) is recognized as an analogue to future projections of a climate system with significantly elevated atmospheric pCO_2 (approximately 400 to 800 ppm [Royer et al., 2012]). Long-term trends of sea surface temperatures (SSTs) through the Late Cretaceous have suggested a global cooling from a Cenomanian (94 to 100 Ma) thermal maximum through the Cretaceous-Paleogene boundary (KPB) [Clarke and Jenkyns, 1999; Miller et al., 2005; Linnert et al., 2014]. Recent work from records of dinoflagellate and terrestrial floral assemblages, TEX_{86} temperature reconstructions, and benthic foraminfera $\delta^{18}O$ values has hinted at the possibility of warm intervals, and/or oscillatory climate variability (3 to 5 °C) at timescales of less than 1 Ma in the Maastrichtian [Bowman et al., 2013; Linnert et al., 2014].

We utilized carbonate clumped isotope paleothermometry to reconstruct SSTs throughout several fossiliferous sections spanning the Late Cretaceous from the southeast US. We analyzed shell material from the Peedee (South Carolina), Ripley (Alabama), Prairie Bluff (Alabama), and Owl Creek (Tennessee) formations across the Mississippi Embayment and the modern Atlantic coastline. We found paleotemperatures ranging from 11 to 25 °C for these localities with strong reproducibility between analyzed taxa representing six genera. These results reveal temperature variations up to 7 °C throughout the Late Campanian and Maastrichtian (~76-66 Ma). There is also agreement between the Prairie Bluff and Peedee Formations, which both record the warmest temperatures just before the KPB, suggesting larger-scale regional connections between these two sites.

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A reconstruction of temperature and $\delta^{18}O$ data for the Chinese Loess Plateau at the Last Glacial Maximum using carbonate soils and snails

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The East Asian Monsoon, an important meteorological phenomenon, supplies a quarter of the world's population with torrential summer rains. In order to investigate how it responds to insolation, deglaciation, and changing levels of greenhouse gases, we are reconstructing temperature and $\delta^{18}O$ using paired clumped isotope and oxygen isotope analyses of carbonates. We are measuring a suite of soil carbonate nodules and gastropods—both modern and ancient—from the Last Glacial Maximum (LGM), around 18,000-23,000 years before the present. Samples are from the Chinese Loess Plateau (CLP), a large region in mid-China bordered by the Qinling and Lüliang Mountains that is on the northern boundary of the East Asian Monsoon. One study [*Eagle et al.*, 2013] has already examined data for a single location on the CLP, which yielded encouraging results. We sampled soil nodules and gastropods from 15 sites in roughly gridded transects across the CLP, which will allow us to investigate spatial variations in temperature and $\delta^{18}O$ and characterize the underlying causes of changes in East Asian Monsoon intensity since the LGM.

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The Δ_{47} fractionation of different carbonate minerals digested with phosphoric acid at 70 °C in a Kiel IV – MAT 253 system

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The broader paleoclimate community is increasingly convinced that the clumped isotope signature (Δ_{47}) in carbonate minerals may be the ideal proxy for reconstructing the formation temperature of carbonates, due to its independence of the isotopic composition of the ambient water in which the carbonate mineral formed. Yet, inter-laboratory discrepancies in Δ_{47}-temperature calibrations and in the measurements of carbonate standards cast a cloud over the application of the Δ_{47} signature. One potential reason for inter laboratory discrepancies could be differences in the isotopic fractionation during phosphoric acid digestion of the carbonate molecule ($^{13}C^{18}O^{16}O_2^{2-}$) to gaseous CO_2 ($^{13}C^{18}O^{16}O$) (Ghosh et al., 2006) caused by differences in sample preparation or the instrumental setup.

In this study we determined the acid fractionation of different carbonate minerals digested at 70 °C in a Kiel IV carbonate device coupled to a MAT 253 mass spectrometer setup (Thermo Scientific, Bremen, Germany). The carbonate minerals were heated for several hours to temperatures of approximately 1000 °C to reach stochastic isotope distribution. Heating of calcites was performed in a "Paterson apparatus", because it enables heating of gram-size carbonate quantities, but has a relatively slow cooling to room temperatures of 45 minutes. Dolomite and siderite were heated in smaller quantities in a "piston cylinder", which allows to reach the higher pressure necessary to stabilize the specific carbonate mineral and ensures rapid cooling within 30 sec. XRD measurements verified the purity of the heated carbonate mineral. Subsequently the carbonates were finely ground, dissolved with phosphoric acid at 70 °C in the Kiel IV and the evolved CO_2 gas analyzed for its mass 44 to 49 in the MAT 253. The intensities were pressure baseline corrected, converted into the absolute reference frame and projected to acid digestion at 25 °C (Meckler et al., 2014).

Preliminary results of the Δ_{47} fractionation for carbonate digestion with phosphoric acid at 70 °C in a Kiel IV device are consistent with results from other studies using different preparation methods (Guo et al., 2009; Defliese et al., 2015). Thereof the various carbonate mineralogies indicate small but significant differences in their acid fractionation, which has to be taken into account while interpreting their Δ_{47} signature.

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Clumped isotopes in shell midden sclerochronology: Prehistoric cooking versus climate signatures?

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Incrementally banded calcareous components derived from prehistoric shell midden deposits like bivalve shells or fish otoliths provide high-resolution records of multiple paleoenvironmental proxies (e.g. δ^{18}OCarbonate or element/Ca ratios). Thus, allowing the reconstruction of sub-seasonally resolved environmental conditions, such records became valuable tools to study ancient climate change. However, differing from other common marine paleoenvironmental proxy archives, shell middens were accumulated by the coastal populations, gathering local faunal resources for dietary purposes. Depending on the available technology and regional customs, the midden's constituents could have been exposed to prehistoric cooking prior to deposition. The alteration of such proxy records by prehistoric cooking has poorly been studied so far and is neglected in most paleoenvironmental reconstruction using shell midden constituents.

Here we present experimentally determined clumped isotope, conventional oxygen and carbon isotope as well as element/Calcium data measured in bivalve shells of the hard clam *Mercenaria campechiensis* exposed to different prehistoric cooking methods. Our data clearly show that pre-depositional heating (i.e. cooking) of aragonitic shells can lead to a considerable alteration of most paleoenvironmental proxies including their clumped isotope signature, even without an apparent reordering of the initial aragonite into secondary calcite. Thus, common methods for assessing the preservation of aragonitic skeletal components (e.g. x-ray diffraction analysis) might fail to detect their potential alteration due to the exposure to pre-depositional cooking methods. However, our data show that clumped isotope thermometry represents a suitable tool for the detection of such pre-depositional heating events and the potential subsequent alteration of paleoenvironmental proxy records. In addition, clumped isotope thermometry also allows the differentiation between certain prehistoric cooking methods and thus appears to be a suitable approach for tracing changes in ancient cooking practices as a consequence of technological advancement throughout human history using shell midden deposits.

Sulfate Reduction and Diagenesis – Possible Disequilibrium in Clumped Isotopes of Dolomitized Tertiary Sediments

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In this study we present clumped isotope data from multiple sites in the Bahamas with varying degress of dolomitization and varying diagenetic regimes as dictated by the depth, location on the platform, and abundance of sulfate reduction. Because of the young-age of the samples being studied and the inherent tectonic stability and consistent fluid temperatures of the Bahamas, the range of possible temperatures exerted on these samples is well constrained. This unique set of circumstances makes these samples a testing ground for clumped isotope paleothermometry on diagenetically altered substrates and dolomitized sediments.

Samples in this study originated from three seperate cores which span the Late Miocene to Pleistocene. The first is from a 168m deep core from the island of San Salvador on the eastern edge of the Bahamas platform with a 120 block in the middle of the core which is completely dolomitized with near stoichiometric to stoichiometric dolomite. The other two cores, Unda and Clino, are from the top of the western edge of Great Bahamas Bank with Unda located 8.5km farther onto the platform. Unda and Clino, at ≈450 and ≈670 m in the length respectively, represent the deepening water environment of the shelf to slope transition. Unda has large sections below 100m depth down core which are completely dolomitized and others that are partially dolomitized, while Clino has limited dolomitization with small sections associated with non-depositional surfaces reaching >80% dolomite, but the vast majority of the core is <40% dolomite.

Both clumped isotopes and $\delta^{34}S$ from carbonate associated sulfate (CAS) were measured on all three of these cores with a focus on the dolomitized and partially dolomitized sections. Samples from both San Salvador and Unda returned temperatures that were within a reasonably expected range of values (San Salvador: 20-30 °C; Unda: 15-23°C). Clino on the other hand consistently returned temperatures that exceeded reasonable formation temperatures in the deeper slope water environment and most were warmer than the possible surface waters where the composing carbonates derived from (>35°C). By chemically treating the Clino samples, the dolomite was isolated and remeasured for clumped isotopes producing similar warm temperatures. This suggests the problem is not solely within the dolomites but associated with the bulk material.

Within carbonates $\delta^{34}S$ from CAS is directly related to the sulfate composition of the fluid. Sulfate in unrestricted ocean waters is principally controlled by the amount of sulfate reduction taking place, is fairly homogenous, and well constrained through the tertiary. Measurement of $\delta^{34}S$ in Unda and San Salvador showed little deviation from expected values during the timing of deposition and dolomitization suggesting little stagnation in the fluid flow throughout the sediments. The bulk samples from Clino however showed large variations (>10‰) from seawater sulfate values. Comparing sulfate values with clumped isotope Δ_{47} suggests a relationship between increasing $\delta^{34}S$ and decreasing Δ_{47}.

Long Term Comparison of Data Collected using Pressure Base Line Correction and Conventional Processing Methods

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Sample Δ_{47} values are most commonly placed into the absolute reference frame (ARF) using a suite of heated and water equilibrated gases to track and correct for changes in the negative pressure baseline using the methods described in Dennis et al. (2011). An additional correction scheme described in He, Olack and Colman (2012) suggested that the negative pressure baseline could be better corrected for by physically measuring its value using off-peak measurements and retroactively adding it back onto on-peak measurements. This process came to be known as the PBL correction and was shown to reduce the dependency of processed Δ_{47} values on the results of heated and water equilibrated gases, however the method often required extending the length of time to measure a sample and produce similar statistical accuracy.

In order to compare the two processing techniques, clumped isotope data has been collected over the course of 12 months using the measurement techniques required to perform the PBL correction and all data post-processed with and without its application. On a subset of this dataset, we have shown that the two techniques produce nearly identical Δ_{47} values with a slight tendency for the PBL value to give more positive values ($\approx 0.01 \pm 0.05$‰). The standard error associated with each Δ_{47} measurement is also often indistinguishable between the two techniques. However, when examining the measurements of a Carrara marble which is measured almost daily in the UM laboratory, it was found that the PBL tended to produce more stable measurements over time. In periods where Carrara Δ_{47} values would go to extreme deviations from the mean using only the ARF, the PBL tended to yield Δ_{47} values closer to the mean. This resulted in standard deviations for the PBL processed Carrara samples that were almost 50 % of the ARF processed values (0.029‰ vs. 0.045‰).

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Baseline modelling.

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The standard way of processing results from clumped isotope studies is the absolute reference frame [Dennis, *et al*., *2011*]. This follows on the work of Huntington, *et al*. [*2009*] where Δ_{47} was plotted *vs*. δ_{47} and the reported Δ_{47} was corrected to δ_{47}=0. He *et al*. [*2012*] showed that the slope of the Δ_{47} *vs*. δ_{47} was a function of shifted baseline in the *m/z* 47 reading, and that the Δ_{47} *vs*. δ_{47} function is not a true line but an arc. Measuring and correcting for the baseline offset minimized the slope, as well as deviation from a linear fit, and data from multiple equilibrated and heated (1000 °C) gases could be combined to calculate said slope. He *et al*. [*2012*] measures the baseline in every run, and others, e.g. Bernasconi *et al*., [2013] determine it periodically. Here we report using a model system to fit the residual baseline to the EG and HG data. That baseline can then be added to the measured baseline, either as a function of the measured baseline reading or the *m/z* 44 reading, during the calculation of the Δ_{47} value. The effective Δ_{47} vs δ_{47} slope should be zero, so the baseline correction does the correction to δ_{47}=0.

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The effects of Porapak^{TM} trap temperature on $\delta^{18}O$, $\delta^{13}C$, and Δ_{47} in preparing samples for clumped isotope analysis

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Preparation of CO_2 for clumped isotope analysis requires additional cleaning steps beyond those used for traditional stable isotope measurements ($\delta^{18}O$, $\delta^{13}C$) because the clumped isotope measurement is susceptible to isobaric interference from a wider range of contaminants, including hydrocarbons, halocarbons, and sulfur compounds [*Eiler and Schauble*, 2006]. To reduce the effects of these contaminants, the CO_2 is cleaned prior to sample introduction into the mass spectrometer in one of two ways: by passing CO_2 through either 1) a gas chromatography column with a helium carrier gas [*Ghosh et al*., 2006 and others] or 2) a "static" U-trap filled with a similar adsorbing agent (such as Porapak^{TM} Q, or "PPQ") using the pressure gradient produced by liquid nitrogen to facilitate the transfer [*Dennis and Schrag*, 2010]. Of the labs using the static PPQ trap configuration, the dimensions of the trap, the temperature to which it is cooled, and the duration of transfer have all been empirically determined on a lab-by-lab basis, possibly through assessing yield (% of starting gas collected on the far side of the PPQ trap) or by testing measured Δ_{47} values of standards calibrated by other labs.

We tested the effects of PPQ trap temperature on the measured stable isotopic and clumped isotopic composition of carbonate standards. We found that, on the University of Michigan sample preparation line, at PPQ trap temperatures below -20°C, $\delta^{13}C$, $\delta^{18}O$ and raw Δ_{47} are fractionated, with the magnitude of the fractionation increasing linearly as temperature decreases below -20°C. The rate of fractionation is 0.0097 ‰/°C below -20°C for $\delta^{13}C$, 0.0392 ‰/°C below -20°C for $\delta^{18}O$, and 0.0037 ± 0.0019 ‰/°C for raw Δ_{47}, in all cases making the measured values lighter at colder temperatures. Reference-frame-corrected Δ_{47} values ($\Delta_{47\text{-RFAC}}$) were unaffected as long as standards gases and samples were prepared identically. Contaminant removal, determined by monitoring Δ_{48}, was equivalent at all PPQ trap temperatures between -10°C and -40°C. Interestingly, measured yield through the PPQ trap was consistently >99% at PPQ trap temperatures as low as -30°C, resulting in a temperature interval (-20°C to -30°C) in which high yields were achieved but fractionation was occurring. This suggests that an ideal PPQ trap temperature cannot be determined by monitoring yield alone (or $\Delta_{47\text{-RFAC}}$). We find that a PPQ trap temperature above -20°C is required to avoid fractionation, and -10°C to -15°C is ideal for the University of Michigan apparatus. Other labs should determine their own ideal PPQ trap temperature by monitoring not only yield and $\Delta_{47\text{-RFAC}}$, but also changes in $\delta^{13}C$ and $\delta^{18}O$ with temperature.

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End-Cretaceous extinction in Antarctica linked to both Deccan Traps volcanism and bolide impact

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The cause of the end-Cretaceous (KP) mass extinction is still under debate. Some suggest that the eruption of the massive Deccan Traps volcanic province in India caused species extinction through negative effects of volatiles emitted during the eruption (e.g. CO_2, SO_2) [*Keller*, 2012]. Others cite the impact of the massive Chicxulub meteorite as the cause [*Alvarez et al*., 1980]. Recent work has suggested that the impactor triggered and accelerated eruption of the Deccan Traps [*Richards et al*., 2015], further complicating the cause-effect relationship.

We present a record of high latitude ocean temperature and $\delta^{18}O_w$ from Seymour Island that covers the last few million years of the Cretaceous and crosses the KP boundary (~69-65.5 Ma). Seymour Island, Antarctica (64°17'S, 56°45'W) is particularly well-suited to studying the KP boundary interval due to its expanded section, continuous sedimentation, and abundant, exceptionally preserved macrofossils, including some species and genera that survive across the KP boundary [*Macellari*, 1988; *Zinsmeister and Macellari*, 1988]. We find an ~8°C warming prior to the KP boundary that aligns with the onset of Deccan Traps volcanism and a pre-KP extinction event identified by [*Tobin et al*., 2012], directly tying extinction at this site to the volcanic trigger event through climate change. A second extinction event at the KP boundary coincides with a second warming pulse, the time of meteorite impact, and additional Deccan Traps volcanism.

The bivalve shells were each measured at two positions: the umbo (hinge) and the ventral margin (back edge). We consistently find large differences in measured parameters between the two positions on the same shell (>5°C, many ‰ in $\delta^{13}C$ and $\delta^{18}O_w$). We suggest these differences are due to different seasonal aliasing early and late in the life cycle of the bivalve. In addition, the coldest temperatures recorded suggest the presence of sea ice and potentially continental glaciers at higher elevations, which is contrary to most assumptions of the Cretaceous as an ice-free world.

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Kinetics of aragonite clumped isotope reordering

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Clumped isotope thermometry potentially constrains the growth temperatures of carbonate phases, simple molecular gases, and perhaps other compounds. Aragonite is an attractive target for this technique because it is a primary precipitating mineral out of the modern ocean, and its presence in ancient sediments usually is interpreted as evidence for little or no diagenetic alteration. Here we examine the kinetics of isotopic re-ordering in aragonite in response to heating, both with and without transformation of metastable aragonite to stable calcite. We conducted a series of heating experiments in which shards of aragonite single crystals were heated under a head-space of CO_2 sealed in pyrex tubes. The proportions of aragonite and calcite were measured by XRD and then analyzed for their stable, including clumped, isotope composition. The results from these experiments show a complex process of both isotopic reordering and phase transformation during heating: In the temperature range where reactions are fast enough to be observed on laboratory time scales (300 ˚C$\leq$ T $\leq$450 °C) samples evolve in three stages: (1) aragonite first undergoes isotopic re-ordering toward equilibrium at the high experimental temperature, decreasing sharply in Δ_{47}; (2) then aragonite begins to transform to calcite, accompanied by an increase in Δ_{47}, recovering most of the Δ_{47} drop that occurred in step 1; note Δ_{47} values never exceed their value before any heating; finally (3) the newly formed calcite decreases in Δ_{47} toward the high temperature equilibrium, at a rate consistent with previous heating experiments performed on calcite [*Henkes et al*., 2014; *Stolper and Eiler*, 2015]. The apparent blocking temperature given typical geologic cooling rates, is approximately 100°C. This indicates that for samples that have undergone little or no diagenetic alteration it is possible to preserve the growth temperature of aragonite, though only with quite shallow maximum burial depths (est. $\leq$ 2-3 km). In this sense, aragonite is more susceptible to thermally activated isotopic reordering than is calcite or (especially) dolomite, and is similar to apatite. Step 2 presents a peculiar and puzzling finding: Δ_{47} values that rise toward a low apparent temperature during sustained heating at high temperature. We suggest this result can be interpreted using the 'pairs' model of Stolper and Eiler [2015], wherein the first step to isotopic re-ordering during heating is disproportionation of 'clumps' to form pairs of adjacent singly substituted carbonate ion units. Our results could be explained if the aragonite to calcite transformation involves selective re-combination of the C—O bonds that were broken during that pair forming reaction. We are currently conducting further experiments to test this hypothesis, which predicts specific changes in behavior for different durations of the initial heating of aragonite.

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Experiments constraining blocking temperatures of H isotope exchange in propane and ethane

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Hydrogen isotope compositions of n-alkanes, including intermolecular fractionations, site-specific intramolecular fractionations, and ^{13}C-D or D-D 'clumping', can potentially provide geothermometers that constrain the temperature histories of petroleum systems (i.e., conditions of hydrocarbon formation, migration, and/or storage). The ability of such fractionations to record temperature depends critically on the 'blocking temperatures' of relevant exchange processes. Prior work on this problem has focused on temperatures far in excess of natural petroleum systems, and/or on mixed materials (oils, kerogens) where it is unclear what species are or are not exchangeable. Here we experimentally constrain the rates of hydrogen isotope exchange at low temperatures (down to 100 ˚C) between ethane, propane and water, with or without catalysts, including experiments that examine whether rates of exchange differ between non-equivalent sites in propane. A motivation for studying the kinetics of exchange at this level of detail is that we speculate blocking temperatures of different compounds and sites may differ from one another, such that the H isotope distributions between and within the alkanes may record several events or conditions.

We performed experiments heating propane and ethane vapor exposed to deuterated water (δD=1,1419 ‰), either alone or in the presence of Ni catalyst or clay, at temperatures between 100 and 400 ˚C. Ethane is not exchangeable at time scales of weeks in the presence of water alone at temperatures up to 400 ˚C, nor in the presence of Ni or clay catalysts at 100 or 160 ˚C, but has an exchange rate of ~10^{-4} hr^{-1} in the presence of Ni at 200 ˚C. Similarly, propane is not exchangeable in the presence of water alone at 200 ˚C or below, nor in the presence of catalysts at 160˚C or below, but exchanges vigorously in the presence of Ni at 200 ˚C (~10^{-3} hr^{-1}). Under these conditions hydrogen isotope exchange of the central CH_2 group of propane is significantly faster (~2x) than that of the terminal CH_3 group. Propane exchange in the presence of clay at 200 ˚C is measurable for the CH_2 group (~10^{-5} hr^{-1}), but not observable for the CH_3 group.

Results from this study indicate that hydrogen exchange with water can control the hydrogen isotope contents of alkanes even at the low temperatures of natural petroleum generating systems. In the presence of Ni catalysts, both ethane and propane are sufficiently exchangeable that geological blocking temperatures should be less than 100 ˚C; at least the central site of propane is also likely exchangeable at low temperatures over geological time scales in the presence of clay. These findings have important implications for the possible uses of inter- and intramolecular fractionations involving natural hydrocarbons as a large family of geothermometers. But they highlight the essential role of catalysis in controlling the blocking temperatures of these systems. Our ongoing work on this problem focuses on understanding the activity of more naturally relevant catalysts in the T and P range of natural systems.

Clumped isotope analyses of sub-Arctic early Cretaceous fossils: implications for seawater isotope values

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Here we present estimates of sub-Arctic marine temperatures and seawater isotope values obtained from Cretaceous fossil mollusk shells using the 'carbonate clumped isotope' method. The well-preserved Early Cretaceous (Valanginian) mollusk shells (belemnites) were derived from a number of high latitude locations in Siberia. From these analyses we infer average marine temperatures of 10-26°C. Our data are hence consistent with a warm 'greenhouse' world and near ice-free polar conditions. Our combined temperature and belemnite oxygen isotope data imply seawater isotope values that have a remarkably modern character in that they are similar to modern high-latitude seawater. We consider the seawater oxygen isotope values to be plausible and point to unexpected basin- or global-scale hydrologies. Our findings are consistent with modelling results that also suggest Cretaceous surface seawater oxygen isotope values are similar in many respects to modern.

253 Plus - New 10 kV isotope ratio mass spectrometer: Robust precision and accuracy in clumped isotope research

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The new ThermoFisherScientific 253 Plus isotope ratio mass spectrometer with proven 10-kV technology offers the highest sensitivity, best peak shape and best peak stability of any commercially available IRMS. The clumped isotope research community requires these essential capabilities for highly reproducible measurement of very small isotope signatures which are related with small isotope abundances. Since more than 10 years the 10-kV technology of the 253 series is successfully used and the instrument of choice for required smallest sample amount analysis in clumped isotope research.

The 253 Plus with its modified magnet design and optimized collector arrays removes the pressure baseline effect of interferences on small isotope signals, e.g. the Δ_{47} –CO2 linearity effect. The new magnet will give uncompromised peak flatness on clumped isotopologues. For even smaller isotopologue signal intensities, the baseline monitoring is useful to correct for second order baseline effects. Additionally, the improved ion transmission enables the highest isotope ratio stability on longer integration times for any dual inlet system measurements. Due to higher throughput and lower sample consumption, the long-integration time dual inlet methodology (LIDI) opens the field of application to smaller sample sizes and thus higher resolution studies.

Furthermore, the excellent noise behavior of the 253 Plus allows implementation of the 1013 ohm technology developed by Thermo Scientific. An additional improvement to sample consumption and reduced sample size use will be given by the combination of the LIDI measurement mode and the 10^{13} ohm technology. The technical details of the new 253 Plus and their influences on long-time robustness in Δ_{47} precision and accuracy of standards and samples will be presented.

Apparent negative clumped isotope signatures

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High precision measurements of molecules containing more than one heavy isotope in environmental samples are becoming available with new instrumentation and may provide novel constraints on element cycles in nature. These so-called clumped isotope signatures are usually reported relative to a random (stochastic) distribution of heavy isotopes over all available isotopocules of a molecule. When multiple indistinguishable atoms of the same element are present at different positions in a molecule, calculating this stochastic distribution usually involves arithmetic averaging over the isotope ratios of all contributing atoms (i.e. at all relevant positions). We show here that this arithmetic averaging always leads to negative apparent clumped isotope anomalies for heavy isotope clumping of indistinguishable atoms. The size of the signal is closely related to the standard deviation of the individual isotope ratios. Such statistical clumped isotope anomalies must occur in any system where two or more indistinguishable atoms of the same element but with different isotopic composition combine in a molecule. Thus they have to be taken into account in data interpretation. Furthermore, they may allow assessment of the heterogeneity of the isotopic pools of atoms that are the substrate for formation of molecules.

Dynamic recrystallization of calcite marbles can reset Δ_{47} values

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Exhumation and accompanying retrograde metamorphism alters the compositions and textures of metamorphic rocks through deformation, water-rock reactions and closed-system diffusion-controlled processes. Here, we report mm-scale differences in Δ_{47} values between static vs. deformed fabrics of a color-banded calcite marble sample (NAX2), collected from the Naxos metamorphic core-complex (Greece). Δ_{47} values of domains drilled from the sample (n=12) show a distinct bimodal distribution. Low Δ_{47} values (0.364-0.374‰; all values are reported in the absolute reference frame) correspond to an apparent temperature of 260°C and are common in white bands; high Δ_{47} values (0.404-410‰), corresponding to an apparent temperature of 200°C, are common in grey bands. $\delta^{18}O$ and $\delta^{13}C$ values show no significant variation, suggesting a closed-system process. Thin-section analyses show that white bands were statically recrystallized, while grey bands were later dynamically recrystallized. White and grey bands show a homogenous low-Mg calcite composition, indicating that differences in Δ_{47} are not controlled by chemistry or mineralogy. All measured fabrics experienced the same thermal history, as dynamic recrystallization is not prone to shear heating. We found no petrographic or isotopic evidence for dissolution/precipitation processes. Instead, we suggest that the low Δ_{47} values reflect diffusion-controlled isotopic reordering during cooling, whereas high Δ_{47} values reflect isotopic reordering driven by dynamic recrystallization. Δ_{47}-based temperatures of the static fabrics are ~60-80°C higher than commonly observed in slowly cooled metamorphic rocks, and, based on models, would suggest cooling rates of ~10^5 °C Myr^{-1}. This is far beyond the rates achievable by exhumation. We show below that the calcite in these fabrics does not have anomalously slow isotopic reordering rate. We suggest that these rocks experienced a heat-shock during their exhumation history, possibly due to local hydrothermal activity. We attribute high Δ_{47} values to the effect of dynamic recrystallization that enhanced lattice-scale mobility of O, C, or carbonate ion groups. We consider two mechanisms by which deformation may induce such an effect: 1) An intrinsic mechanism by which deformation increases O and C diffusivities in the crystal lattice by increasing defect-density; and 2) an extrinsic mechanism, in which slip along crystal planes efficiently transport carbonate. If the intrinsic mechanism is correct, isotopic re-ordering in dynamic fabrics may remain faster than that in static fabrics (i.e., if differences in defect-density persist). However if the extrinsic mechanism is correct, enhanced mobility occurs only during active strain, and we should find no such difference today. To test these predictions, we compared reaction rates of static and dynamic fabrics in heating experiments. In both fabrics, average Δ_{47} values gradually decrease from initial values, and are consistent with modeled Δ_{47} values based on published rate constants. There is no significant difference between reaction rates in the static and dynamic fabrics, consistent with the extrinsic mechanism. Results suggest that Δ_{47} values of dynamically recrystallized fabrics that form below the diffusion-controlled blocking-temperature for calcite constrain the temperature of deformation. The variation of Δ_{47} values records a complex thermal/deformational history that includes: 1) gradual cooling to below blocking-temperature; 2) heat shock, followed by rapid cooling; and 3) dynamic recrystallization resetting of Δ_{47} values in the shear bands.

Paleotemperature estimation in a semi-enclosed marginal sea in the East Asia

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The great advantage of clumped isotope thermomety of carbonate minerals is no influence of isotopic composition of seawater, in which organisms precipitate their shells, skeletons, etc. It implies that there is a potential to reconstruct past isotopic composition of seawater as well as temperature using the clumped isotope method. This approach may be useful especially for samples precipitated in extreme environment, which conventional thermometries, such as alkenone and Mg/Ca, are not applicable.

The Japan Sea is one of marginal seas locating eastern edge of the Eurasian continent. This sea connects to Pacific Ocean and adjacent marginal seas by narrow and shallow four straits. The shallowest one, Tartar Straits, has sill depth of ~15 meter, and second shallowest Soya Straits is ~55 meter. The remaining two straits, Tsushima and Tsugaru Straits, have sill depth of ~140 meter, which is similar to amplitude of sea-level lowering at the last glacial maximum. According to such geographical settings, the sea is nearly isolated from connecting seas during the glacial periods, and the limited seawater exchange makes surface environment anomalous. The oxygen isotopic composition of planktonic foraminifer shows a remarkable negative peak at the LGM. Because the surface water at that time was thought to be cool as was in other seas, low salinity is supposed to be responsible for the light isotope values. However, the temperature and salinity at LGM have not yet been understood.

We collected sediment cores from southern part of the Japan Sea in order to reconstruct paleotemperature at the LGM. These contain enough amounts of foraminifera in intervals of LGM. We analyze Δ_{47} of fossil foraminifer using MAT253 in Tokyo Institute of Technology. We also analyze a fossil molluscan shell, which is retrieved from seafloor sediment, and its radiocarbon age is around 22,000 years BP. The preparation and calibration methods follow those of Ghosh et al. [2006], Dennis et al. [2011], and He et al. [2012]. Since the measurements are still in progress, the results will be shown in the workshop.

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Is the absolute reference frame absolute? An apparent dependency of Δ_{47} on $\delta^{13}C$ in CO_2

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The absolute reference frame (Dennis et al. 2011) was designed to calibrate all laboratories to the theoretical predictions of the Δ_{47} values of CO_2 gases at thermodynamic equilibrium determined by Wang et al. (2004). Researchers obtain reference frame CO_2 gases at a range of bulk δ_{47} compositions and select temperatures. The measured Δ_{47} values of these gases are used to standardize Δ_{47} values of samples. It seems that most labs obtain the range of bulk δ_{47} compositions needed to build this reference frame by varying the $\delta^{18}O$ of water that is then equilibrated with CO_2 of a set $\delta^{13}C$ and $\delta^{18}O$ composition, such that the $\delta^{13}C$ value of the reference frame gases remains constant. We present two sets of data that together suggest a dependency of Δ_{47} on $\delta^{13}C$. (1) We create two distinct reference frames using two types of CO_2 that differ by 25 ‰ in $\delta^{13}C$ space. If these two references frames are applied to associated carbonate standards, a Δ_{47} difference of 0.06 ‰ is observed between the calculated Δ_{47} values of the carbonates. (2) We show a linear correlation between $\delta^{13}C$ and Δ_{47} values of CO_2 gases that are all equilibrated at 25°C. Even though these gases are all equilibrated at 25 °C, the measured Δ_{47} values vary by 0.1 ‰ and correspond to estimated equilibration temperatures of 15 to 35 °C. Two immediate implications surface: (1) lack of agreement among labs measuring the same material using disparate $\delta^{13}C$-CO_2 reference frames, and (2) within-lab inflated temperature variation when a wide range in sample $\delta^{13}C$ exists. Furthermore, this phenomenon may explain some of the variation in Δ_{47}-temperature calibrations: we show that when sample Δ_{47} values are corrected for their $\delta^{13}C$ composition, Δ_{47}-temperature relationships collapse into better agreement.

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Miocene-Pleistocene paleoclimate and paleoenvironment in the Meade Basin, Kansas

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The Meade Basin in southwestern Kansas preserves a unique record of paleovegetation and small mammal faunal change from the Miocene to the Pleistocene. Many of the paleosols preserved in this basin contain paleosol carbonate nodules, thick calcretes and abundant organic-rich horizons, which makes it ideally suited for a multiproxy study that explores the role of paleoenvironmental change in driving floral and faunal change. Prior work in this region revealed an increase in the $\delta^{13}C$ values of the paleosol carbonates, which is though to reflect an increase in C_4 vegetation in the Great Plains during this time; the driver(s) of this vegetation change remain uncertain however [*Fox et al*., 2012].

To investigate the drivers of ecological change during this time, we characterized paleoenvironmental changes preserved in the Meade Basin sedimentary record using a variety of proxies, including carbonate clumped isotope paleothermometry, compound-specific isotope analysis of organic matter, as well as elemental and magnetic proxies for ancient precipitation from paleosol samples. Here, we focus on the carbonate samples where we measured carbon and oxygen isotopes ($\delta^{13}C_c$ and $\delta^{18}O_c$, respectively); used clumped isotope thermometry (Δ_{47}) to estimate soil temperature and soil water $\delta^{18}O$; and assessed the preservation state and additional paleoenvironmental features of the samples using optical and cathodoluminescence (CL) microscopy. The carbon isotope record matches previous studies from the region and shows an increase in the relative abundance of C_4 biomass on the landscape since the late Miocene. The Δ_{47} temperatures and the $\delta^{18}O$ of soil water, while variable, show no significant change in average values through time. The textural and luminescence characteristics suggest some samples have undergone moderate to extensive diagenetic alteration from groundwater fluids, perhaps causing some of the variability in the geochemical records. Soil depth may also account for some of the variability. Overall, these data suggest that temperature is unlikely to be the dominant factor driving paleovegetation and faunal change in this region from the Miocene to Pleistocene. In addition, these data highlight the importance of assessing preservation for all carbonate samples, regardless of whether or not the samples have been deeply buried.

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Stable and clumped isotope behaviour over the aragonite-calcite transition: implications for sample prep and proxy interpretation

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At the earth's surface, aragonite is thermodynamically unstable and will readily convert to calcite; this can be accelerated by heating and mechanical stress. In experiments where aragonite is directly heated, we have shown that the alteration of the mineral structure shows a linear correlation between changes in the $\delta^{13}C$ and $\delta^{18}O$ of the carbonate. Experiments conducted in sealed vessels showed that the isotopes did not change when heated in a vacuum, and showed a different relationship when exposed to isotopically labeled CO_2, this indicates that the change in mineralogy facilitates isotopic exchange with atmospheric CO_2. Using different gasses, we are able to calculate the isotopic enrichment ($\alpha_{gas\text{-}carbonate}$) of this process. $\delta^{13}C$ displays correlation (r^2 = .99) between phases however $\delta^{18}O$ displays weaker correlation (r^2 = .63), likely due to the release of water during heating.

Clumped isotopes (Δ_{47}) behave very differently from $\delta^{13}C$ and $\delta^{18}O$ during this transition. Even in a vacuum Δ_{47} will change independently from, and invariably at a faster rate than the mineral transition. At temperatures relevant to the mineral transition (>300°C) Δ_{47} rapidly equilibrates to the higher temperature, likely due to solid-state diffusion. This process has been described in other carbonate minerals (Passey and Henkes 2012, Stolper and Eiler 2015) however the Δ_{47} of aragonite appears to be much more sensitive to heating that of Calcite, Dolomite or Apatite. At temperatures less that 300°C, the mineral transition is minimal, however Δ_{47} will continue to alter at temperatures below 175°C.

The heat and stress from drilling samples is sufficient to partially convert aragonite to calcite, previous work has shown a relationship between $\delta^{18}O$ and mineralogy under these conditions (Waite and Swart 2015). The effect of this alteration on the $\delta^{18}O$ proxy will typically result in a small change, usually less than .5‰ (approximately +2°C). We evaluated the behavior of this transition on the Δ_{47} of sclerosponge and coral skeletons. Using a dental drill and an automated micromill we sub-sampled skeletal material using different drill settings. Under some conditions, we were able to convert over 20% of the skeletal material to calcite. The extent of this conversion varied depending on drill RPM and raster speed. The Δ_{47} decreased with the mineralogy, decreasing by .05‰ (approximately +10°C) accompanying a 20% conversion. The existence of a linear relationship between the Δ_{47} and mineral change indicates that the heat produced by the drill is highly localized, completely altering small portions and leaving others unaltered. We therefore recommend that future studies sampling aragonite archives pay close attention to minimizing the mineral shift. Δ_{47} can be corrected if the mineral transition can be quantified (e.g. using XRD) using the empirical correction equation: $\Delta_{47(corrected)} = \Delta_{47(measured)} + .28‰ \times f_{calcite}$.

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Isotopic Exchange with Phosphoric Acid in the Clumped Isotope System

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The clumped isotope system in carbonates (Δ_{47}) relies on the extraction of CO_2 from carbonate minerals using ortho-phosphoric acid. Despite the fact that this method dates back to the original stable isotopic work in the 1950s, there are significant aspects of the fractionation of the $^{18}O/^{16}O$ during this reaction (and by inference the ratio of masses 47 to 44) which are not understood. We believe that subtle variations in the isotopic fractionation as a function of temperature, acid density (and acid preparation method), and perhaps extraction line design cause variation between the clumped isotope data produced by different laboratories. One of the most obvious of these factors is the difference in reaction temperature. While most laboratories employ temperatures of between 75 and 90°C, the original methods for conventional and clumped isotope extraction (Ghosh et al., 2006; McCrea, 1950) employed a temperature of 25°C. Although various estimate of the difference in fractionation of Δ_{47} between 25 and 90°C have been made, we have measured significantly different values for dolomites compared to published data (Murray et al., 2016). In order to understand this we have performed experiments in sealed Pyrex vessels to measure the exchange between CO_2 and 105, 100, and 98% phosphoric acid. We have determined there to be significant and measurable changes in the Δ_{47} of CO_2 when exposed to phosphoric acid. This exchange is a function of temperature, time, acid strength, and the surface area of the acid exposed to the CO_2. We postulate that, perhaps as a result of the lower reaction rate of dolomite, compared to calcite, that there is greater opportunity for CO_2 to exchange with the phosphoric acid as bubbles of CO_2 are retained within the acid for longer periods of time.

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Clumped and oxygen isotope compositions of last interglacial pedogenic carbonates from Dunaszekcső, Hungary

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Pedogenic carbonates (PC) preserve information on past temperatures and $\delta^{18}O$ of soil water from which they were precipitated [*Cerling and Quade*, 1993]. Studies on recent soil carbonates (SC) demonstrate that 1) they form in the summer season, 2) and thus record summer season temperatures and $\delta^{18}O$ of precipitation [*Breecker et al*., 2009; *Passey et al*., 2010; *Peters et al*., 2013; *Quade et al*., 2013], and 3) that the $\delta^{18}O_{sc}$ is not affected by evaporation provided the carbonates were formed at depths >40 cm [*Hough et al*., 2014].

Here we studied the clumped and oxygen isotope compositions of PCs from a last interglacial paleosol in south Hungary, which was formed during MIS 5e to 5c as revealed by post-IR IRSL dating [*Újvári et al*., 2014]. SCs were found in 4 samples (0.00-0.15, 1.05-1.20, 1.20-1.35 and 1.35-1.50 m depths), and the $\delta^{18}O_{sc}$ values ranged from –7.78 to –7.19‰. Δ_{47}-based soil temperatures (ST) calculated with the equation of *Kele et al*., [2015] are in a range of 19.2 to 22.5 ºC for depths >1.05 m, but the deepest (1.35-1.50 m) sample gave a temperature of 4.8 ºC. A cooler temperature (12.5 ºC) was also recorded by an SC sampled at a shallow depth (0.00-0.15 m). In general, the obtained STs are in good agreement with measured recent summer season STs for 50-100 cm depths (21.4-19.1 ºC). $\delta^{18}O$ of the paleo-precipitation, as calculated from paired $T_{\Delta 47\text{-}sc}$ and $\delta^{18}O_{sc}$ measurements range from –7.55 to –6.97‰ in broad agreement with the summer season $\delta^{18}O_{prec}$ values (–6.18±0.94‰) measured at the Zagreb GNIP station between 1980-1995. Obviously, further measurements are needed to check if the lower $T_{\Delta 47\text{-}sc}$ really record low STs or simply result from analytical issues. Nevertheless, disregarding the two lowest STs, these findings imply that PCs were also likely formed in the summer season during the last interglacial.

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Understanding pressure baseline through measurements from an Isoprime 5kV mass spectrometer

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Reliable measurements of clumped isotopes using a prototype Isoprime dual inlet mass spectrometer system have been demonstrated by Rosenheim (et al., 2013), and used for measurements of both siderite (Fernandez et al., 2013) and calcite (Tang et al., 2014). However, the relationship between Δ_{47} and δ^{47} is steeper than most other platforms indicating poor approximation of measurement background in the Isoprime. Here we will present the status of this system after the installation of a new head amplifier; which has allowed us to observe Δ_{47} signals with lower beam intensities and ultimately smaller sample sizes. The ability to make meaningful measurements at smaller beam intensities ultimately allows us to better constrain the differing, and sometimes perplexing, effects of pressure baseline correction at different beam intensities.

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Isotopic Exchange between CO_2-H_2O and CO_2-H_3PO_4

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Previous studies (Mills & Urey, 1940; Affek, 2013; Clog et. al., 2015) have investigated the rates of oxygen isotopic exchange with CO_2 and H_2O, as well as the Δ_{47} for CO_2 with liquid water. The understanding of the kinetics of these reactions has led to knowledge of the ease of equilibration of CO_2 in the presence of water. We analyze samples at higher temperatures (100°C) than most other laboratories, and are thus interested in the cost of higher water vapor pressures associated with faster reaction times afforded by higher temperatures. Questions remain unanswered surrounding the ease of re-equilibration of CO_2 in the presence of liquid water and phosphoric acid in digestion for clumped isotope analyses. We will investigate the kinetics of isotopic exchange between CO_2 and H_2O as well as CO_2 and H_3PO_4 at 0°C, 25°C and 100°C on varying intervals of time. Exchange between stochastic CO_2 and ice (0°C) will be investigated in order to observe the possibility of isotope exchange between a solid and a gas. Such isotope exchange may affect reference frames in different laboratories, contributing to the presence of different sloped calibration lines between laboratories.

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Paleoenvironmental changes at the Middle–Late Jurassic transition: deciphering T and $\delta^{18}O_{water}$ variations from the clumped isotope signal

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Prominent oceanographic and environmental changes are observed at the Middle–Late Jurassic transition. Boreal seas were characterized by continuous siliciclastic sedimentation allowing us to reconstruct climatic variations in this time period.

A prolonged (Late Callovian–Middle Oxfordian) period of the presence of cold (5– 8.5 ºC) bottom waters is suggested for the epicontinental Middle Russian Sea, belonging to the Boreal province, as based on $\delta^{18}O$ values of belemnite rostra [Wierzbowski et al., 2013]. The occurrence of the cold bottom waters is interpreted as a result of the establishing of wide marine connections with the Arctic Sea during a sea-level highstand. The cooling is sometimes correlated with similar but shorter cooling events known from the Western Europe. The cool period is followed by a pronounced Late Oxfordian–earliest Kimmeridgian warming, which is estimated to 6.5–9.5 ºC based on $\delta^{18}O$ values and elemental ratios of belemnite rostra [Wierzbowski et al., 2013]. As the $\delta^{18}O$ record studied may be affected by salinity variations real palaeotemperatures may by deciphered from the clumped isotope composition of well-preserved carbonate fossils.

Well-preservation state of studied fossils from the Russian Platform is confirmed by the preservation of metastable aragonite and low thermal maturity of the organic matter [Wierzbowski et al. 2013; Bushnev et al., 2006]. Clumped isotope analyses were performed on homogenized powder of 4 belemnite rostra (low-magnesium calcite) and 1 ammonite shell (aragonite) derived from distinct stratigraphic levels of the Russian Platform. At least 5 replicates (each ca. 4 mg) were run for all samples. Material was digested at 90 °C in a common acid bath connected to a fully-automated device for cryogenic purification and GC cleaning of CO_2 gas. Analyte gas was measured on a Thermo Scientific MAT 253. $\Delta_{47,raw}$ data correction comprised the application of a background correction scheme [Fiebig et al., 2015] and the direct projection to the absolute scale [Dennis et al., 2011]. The calcite calibration line of Wacker et al. [2014] was used for $T(\Delta_{47})$ estimations.

Preliminary clumped isotope data show a higher temperature (ca. 10 ºC) of bottom waters of the Middle Russian Sea during the Middle Oxfordian, and probable salinity stratification of the water column. It raises doubts about the theory of the pronounced cooling at the Middle–Late Jurassic transition. The clumped isotope data also show that the Late Oxfordian–earliest Kimmeridgian warming of the bottom water was lower than previously assumed and may have amounted to ca. 5 ºC; the rest of the observed $\delta^{18}O$ variations may have been related to a decrease in salinity of the local seawater.

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Calibration of Dolomite Clumped Isotope Thermometry

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Differences between dolomite and calcite clumped isotope calibrations have yet to be resolved empirically across a range of temperatures. Acid fractionation differences calculated by Guo et al. (2009) suggest that dolomite Δ_{47} values should be ~0.02 ‰ lower than calcites formed at the same temperature. Empirical calibrations for siderite (Fernandez et al. 2014) and apatite (Eagle et al. 2010), however, suggest that a universal Δ_{47} – temperature relationship may exist across all carbonate-bearing mineral phases.

We analyzed five synthetic and four natural dolomites formed at known temperatures. Synthetic dolomites were grown in Mg-Ca-Cl solutions at temperatures of 200 – 235 °C. Natural samples are constrained by fluid inclusion analyses (~90 °C), tropical climate (~25 and ~27 °C), and ocean water column depth (~3 °C).

These data result in a calibration line that is statistically indistinguishable from the high acid reaction temperature calibration of Defliese et al. (2015). At least with current measurement capabilities, we find no evidence for a consistent dolomite Δ_{47} offset. We also found no evidence for a relationship between the degree of dolomite cation ordering and Δ_{47}. These results further support the idea of a universal calibration for carbonate clumped isotope thermometry and enable new investigations into conditions of dolomite formation.

Defliese et al. (2015) Compositional and temperature effects of phosphoric acid fractionation on Δ_{47} analysis and implications for discrepant calibrations. *Chemical Geology*, 396, 51-60.

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Automated clumped-isotope analysis of O_2 on a compact, high-resolution gas-source isotope ratio mass spectrometer

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Analysis of clumped isotopes in molecular oxygen is challenging: The low abundance of mass-35 and 36 O_2 isotopologues (1.5 and 4.2 ppm relative to O_2 in air), isobaric interferences arising from ^{35}Cl, $H^{35}Cl$, and ^{36}Ar, and oxygen's propensity to be reordered on impure zeolite surfaces have limited the external reproducibility of previous efforts to between ±0.1‰ to ±0.2‰ in Δ_{36} values (1σ). Applications of the Δ_{36} tracer in atmospheric and biogeochemical oxygen cycling likely require higher precision to be of maximum utility, however. Moreover, as the number of interesting clumped-isotope systems increases, compact, low-cost alternatives to large, expensive, ultra-high-resolution instruments will help the field diversity more quickly.

To this end, we have constructed a new sample preparation and analysis system at Rice University for clumped isotopes. It consists of an automated gas-chromatographic sample preparation line linked to a Nu Instruments *Perspective IS* isotope ratio mass spectrometer. The *Perspective IS* has a specially modified ion source that achieves a resolution suitable for resolving $^{18}O^{18}O$ from its isobars (mass resolving power ≈ 3500). Unlike ultra-high-resolution instruments, however, the modified *Perspective IS* retains much of its sensitivity. These modifications allow rapid, isobar-free determination of Δ_{36} values in 80 μmol O_2 with precision of ±0.05‰, ±4 ppm, ±0.2‰, and ±0.06‰, respectively (1σ), for $\delta^{18}O$, $\Delta^{17}O$, Δ_{35}, and Δ_{36} values in as little as 3 hours start-to-finish. This new method provides a fourfold decrease in preparation-and-analysis time, a threefold improvement in precision for Δ_{36}, and a twofold decrease in minimum sample size.